I225E Statistical Signal Processing

1. Course Introduction/ Review of Probability Theory

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Statistical Signal Processing

Objective

Basic understanding of random variables, probability theory, stochastic process, statistical signal processing, and model estimation theory.

Contents

Definition of stochastic process, stochastic process and system, correlation analysis, spectral analysis, maximum likelihood estimation, least squares estimation, Bayesian estimation, Wiener filter, Kalman filter.

Lecture Slides

https://candyolivia.github.io/teaching/FY2026/I225E/

Pass: Class-I225E

Material

- Review of probability theory
- Basics of stochastic process
- Stochastic process and systems
- Spectral analysis

Modeling part
1st exam

- Estimation theory
- Maximum likelihood estimation
- Least squares algorithm
- Bayesian estimation
- Linear model
- Signal processing (Wiener filter, Kalman filter)

Inference part
2nd exam

Lecture Schedule (1)

1. Course introduction / Review of probability theory **(15 Apr 2026)** **Report 1 ← Today! <Tutorial hour 1>**
2. Definition of stochastic process **(17 Apr 2026)**
3. Basics of stochastic process **(22 Apr 2026)**
4. Stochastic process and system I **(22 Apr 2026)**
5. Stochastic process and systems II **(24 Apr 2026)** **Report 2**
6. Spectral analysis I **(1 May 2026)**

<Tutorial hour 2> (7 May 2026)

<Tutorial hour 3> (8 May 2026)

Lecture Schedule (2)

7. Spectral analysis II **(13 May 2026)**
8. Estimation theory **(13 May 2026)**

Mid-term Examination (15 May 2026)

9. Maximum likelihood estimation **(20 May 2026)**
   <Tutorial hour 4> Report 3
10. Least squares estimation **(22 May 2026)**
11. Bayesian estimation **(27 May 2026) <Tutorial hour 5>**
    Report 4
12. Linear model **(29 May 2026)**
13. Signal processing I **(3 June 2026)**
14. Signal processing II **(3 June 2026) <Tutorial hour 6>**

Final Examination (8 June 2026)

Reports and exams

- **Viewpoint of evaluation:**

  Comprehension of basic theory of stochastic processing and statistical signal processing and their applications in information science.

- **Evaluation criteria:**

  Reports (20%): Four times (each 5%)

  Examinations (80%): Two times (each 40%)

  Bring only your own “handwritten” notebooks

  (NO printed handouts, calculator, textbook, computer, or smartphone).

Related Courses

- I114 Fundamental Mathematics for Information Science
- I119 Statistics for Data Analytics
- I212 Analysis for Information Science
- I213 Discrete Signal Processing
- I225 Statistical Signal Processing
- I439 Speech Information Processing

1-1 COURSE INTRODUCTION

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1. Introduction

What is random variable?

All possible outcomes that may result from a trial are known a priori. However, it is impossible to predict which outcome to occur.

(ex) dice throwing, coin tossing, etc.

What is Probability Theory?

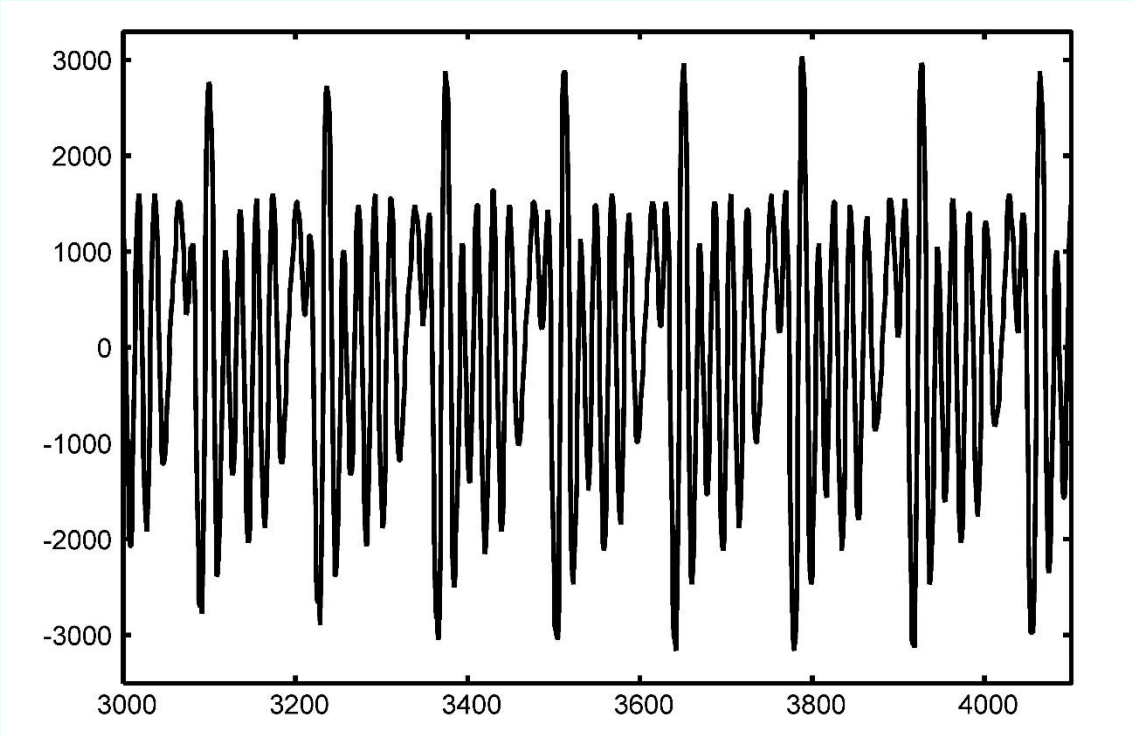
Consider mathematical models for stochastic phenomena, to which only probabilistic predictions can be made.

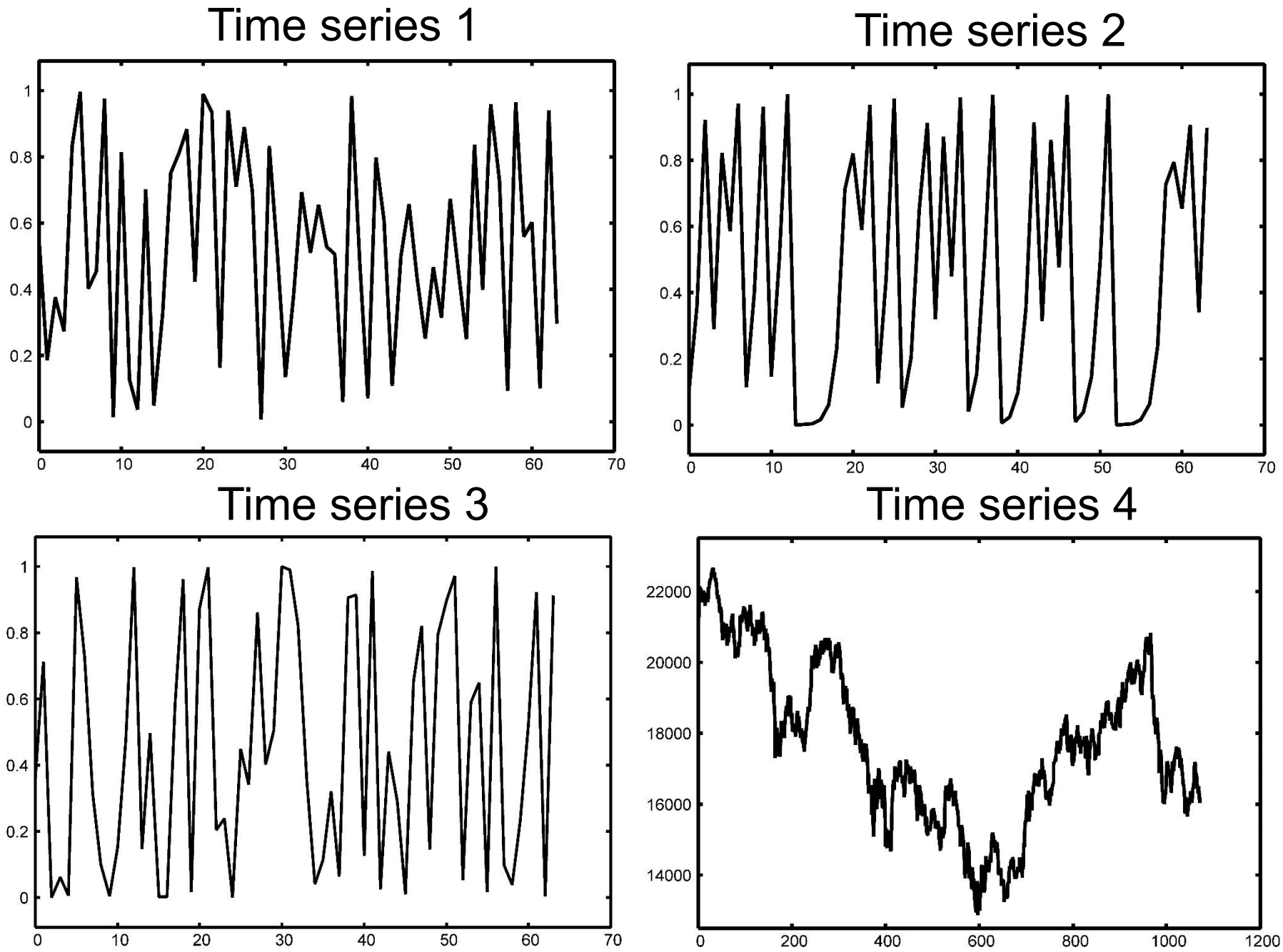
One of mathematical areas.

Foundations by Pascal, Fermat, Huygens already in the 17th century.

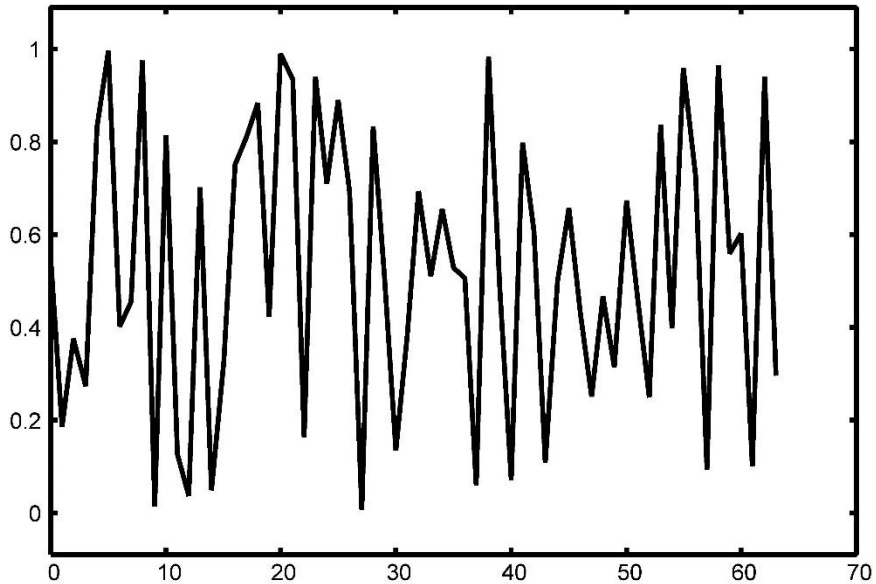
What is Stochastic Process?

Dynamic phenomena that *change* in time in accordance with probabilistic rule. Stock exchange market, price change. Brownian motion (random behavior of particles).

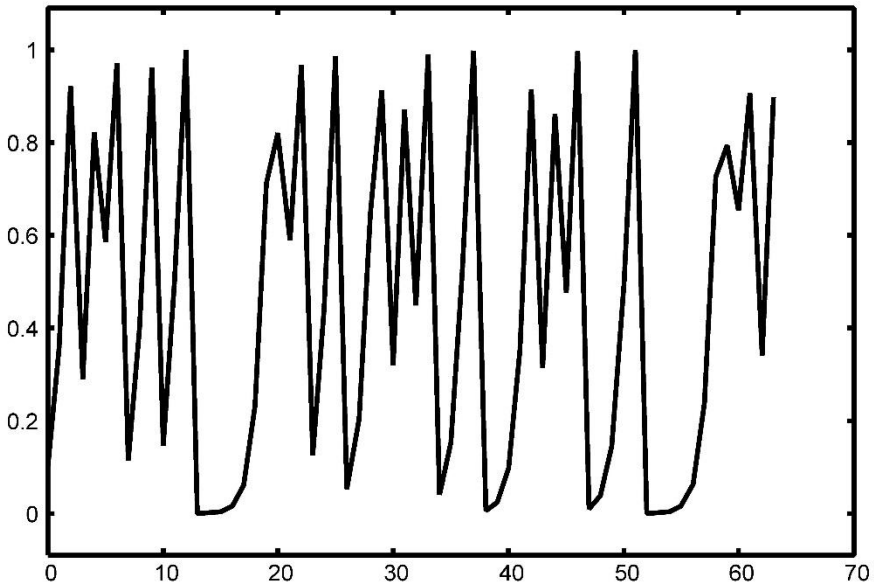



Time series 1
Time series 2
Time series 3
Time series 4
1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
50
60
70
22000
20000
18000
16000
14000
0
200
400
600
800
1000
1200

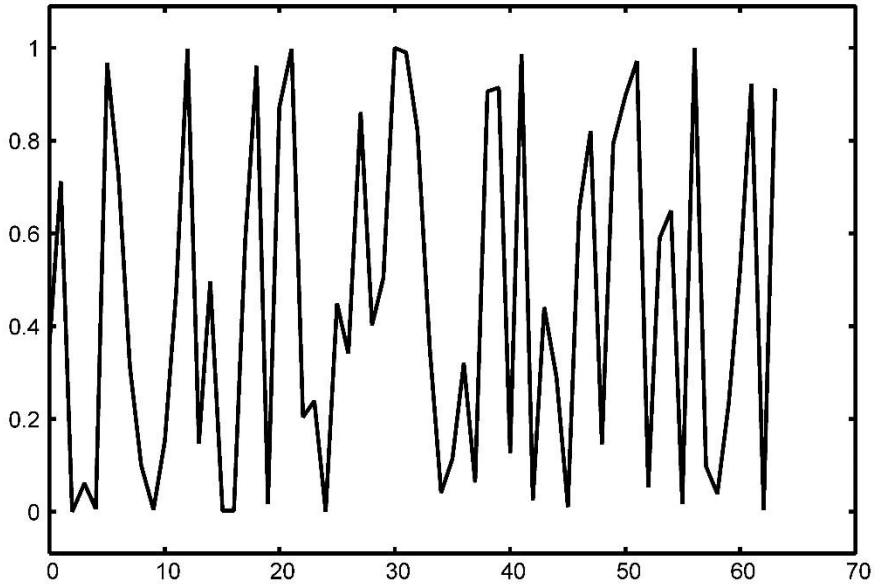
Random numbers



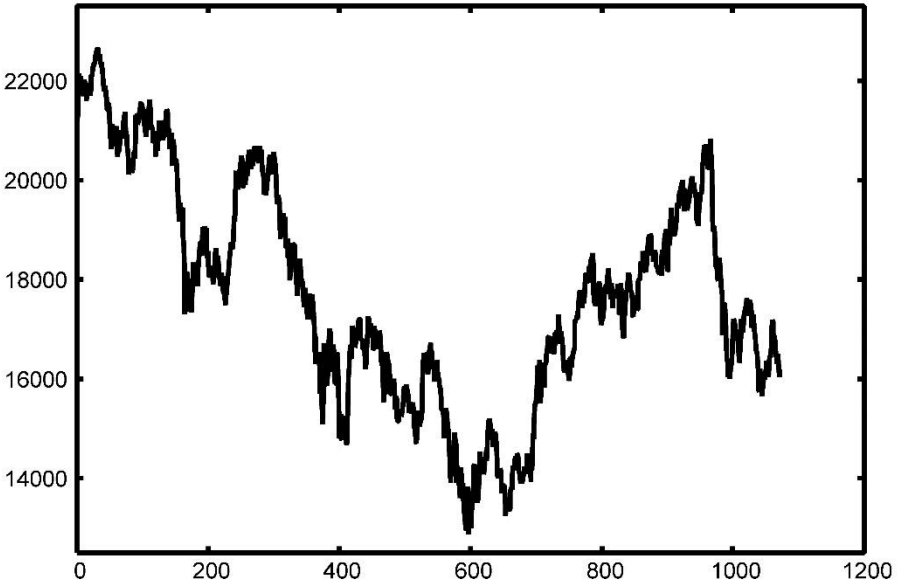
Logistic map



Random shuffling



Nikkei 225 (stock market)



Logistic map

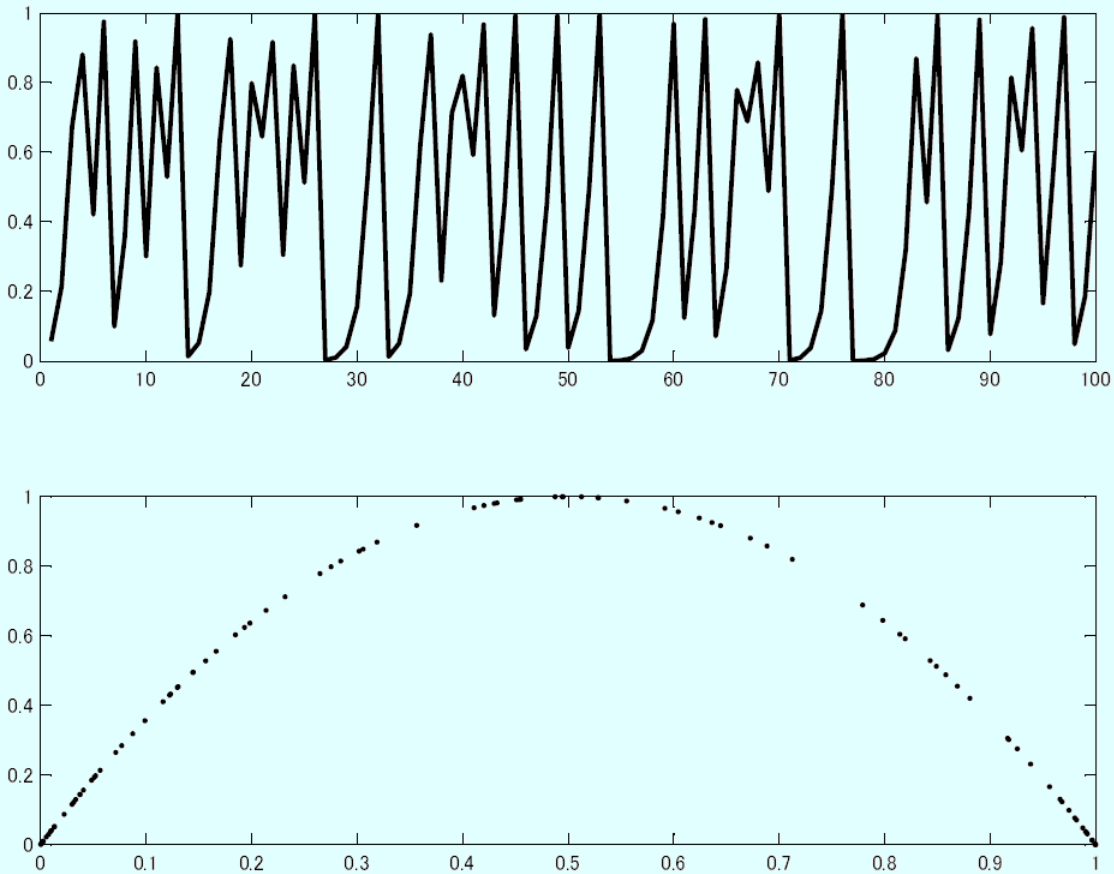
$$x_{n+1} = ax_n\left(1 - x_n\right)$$

```
a=4;

N = 100;
x = zeros(1, N);
x(1) = rand;

for n=1:N-1
   x(n+1) = a*x(n)*(1-x(n));
end

subplot(211);
plot(x, 'k');
subplot(212);
plot(x(1:end-1), x(2:end), 'k.');
```

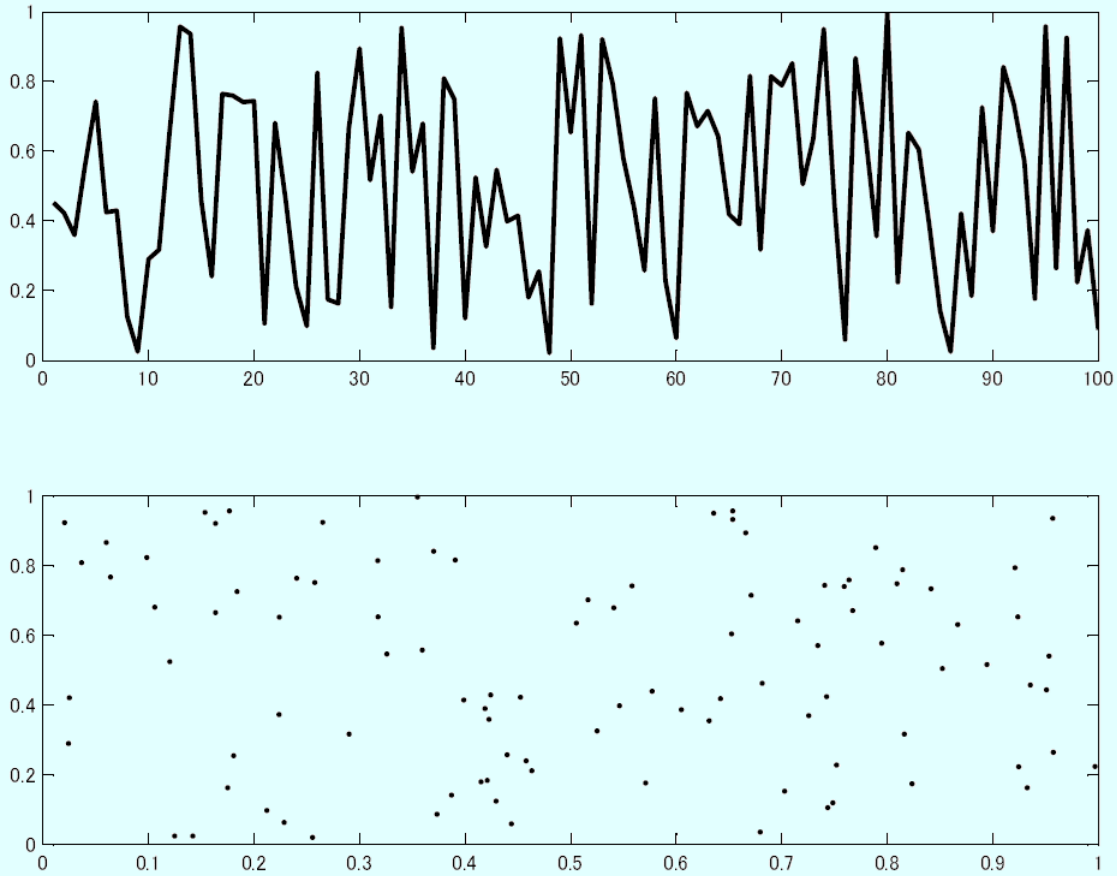


Random numbers

```
N = 100;

x = rand(1,N);

subplot(211);
plot(x, 'k', 'linewidth', 2);
subplot(212);
plot(x(1:end-1), x(2:end), 'k.');
```

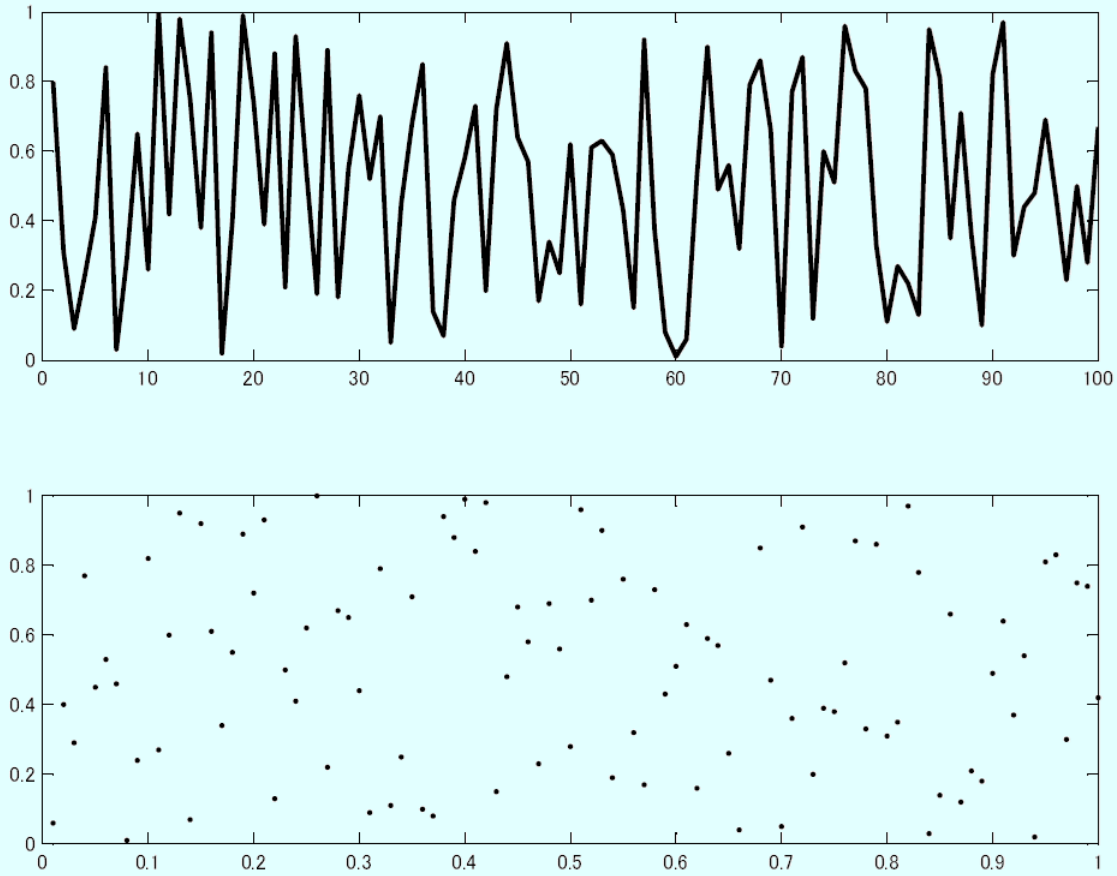


Random shuffling

```
N = 100;

x = randperm(N)/N;

subplot(211);
plot(x, 'k');
subplot(212);
plot(x(1:end-1), x(2:end), 'k.');
```



Random variables and stochastic process

- Random variable
  - History independence; independence of previous values

$$X \sim P(X)$$

- Stochastic process
  - History dependence; current value depends on previous values.

$$X_t \sim P(X_t | X_{t-1}, X_{t-2}, \cdots)$$

  - Markov process

$$X_t \sim P(X_t | X_{t-1})$$

Application of Statistical Signal Processing

- **Time Series Analysis**
  - Modeling
  - System identification (parameter estimation)
  - State estimation
  - Prediction
  - Control
- **Statistics**
  - Basic statistical analysis
  - Statistical estimation
  - Statistical test

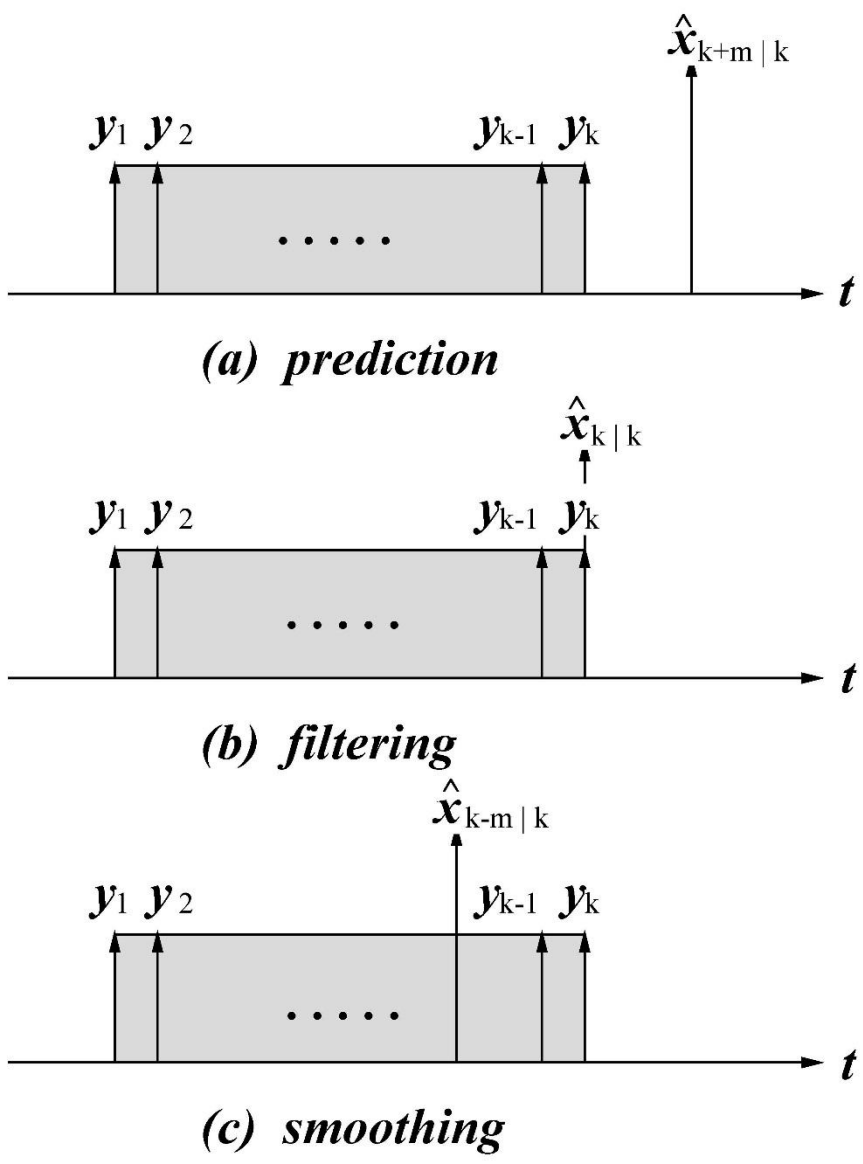
- **Other applications**
  - Information theory
  - Filter theory
  - Secure communication
  - Data coding
  - Speech/Image restoration
  - Machine learning

Time Series Analysis

- **Characterization of Time Series $x(t)$**
  - Mean (First-order Moment), Trend
  - Autocorrelation (Second-order Moment)
  - Spectrum
- **Estimation of System Dynamics**
  - Differential Equation, Linear System
- **State Estimation**
  - Past (Smoothing, Noise Reduction)
  - Present (Filtering)
  - Future (Prediction)

- **Prediction (Future)**
- **Filtering (Present)**
- **Smoothing (Past)**

  Noise Reduction



1-2 Review of Probability Theory

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Review of probability theory

- Probability theory
  Sample space, Borel set, conditional probability, Bayes' theorem
- Random variables
- Distribution functions, density functions
- Joint distributions
- Moments
- Characteristic functions
- Law of large numbers
- Central limit theorem

1. Review of Probability

All possible outcomes that may result from a trial (experiment or observation) are known ***a priori***.

However, it is impossible to predict which outcome to occur.

- Trial $\boldsymbol{S}$: Doing experiment or observation
- Sample Point ω: Individual outcome that results from each trial
- Sample Space $\boldsymbol{\Omega}$: All sets of sample points
- Event $\boldsymbol{A}$: Subset of sample space

When $\omega \in \boldsymbol{\Omega}$ for $\omega \in \boldsymbol{A}$, we say event A took place.

[Example]

- Trial ***S***: Throw a dice
- Sample Point ω : 1, 2, 3, …
- Sample Space $\boldsymbol{\Omega}$: {1, 2, 3, 4, 5, 6}
- Event ***A***: Odd {1,3,5} and Even {2,4,6}

- Trial ***S***: Twice coin-toss
- Sample Point ω : head (h) or tail (t)
- Sample Space $\boldsymbol{\Omega}$: {hh, ht, th, tt}
- Event ***A***: only one head showed

Event

(E1) Complementary Event: $\boldsymbol{A}^C = \{\omega \in \boldsymbol{\Omega}: \omega \notin \boldsymbol{A}\}$

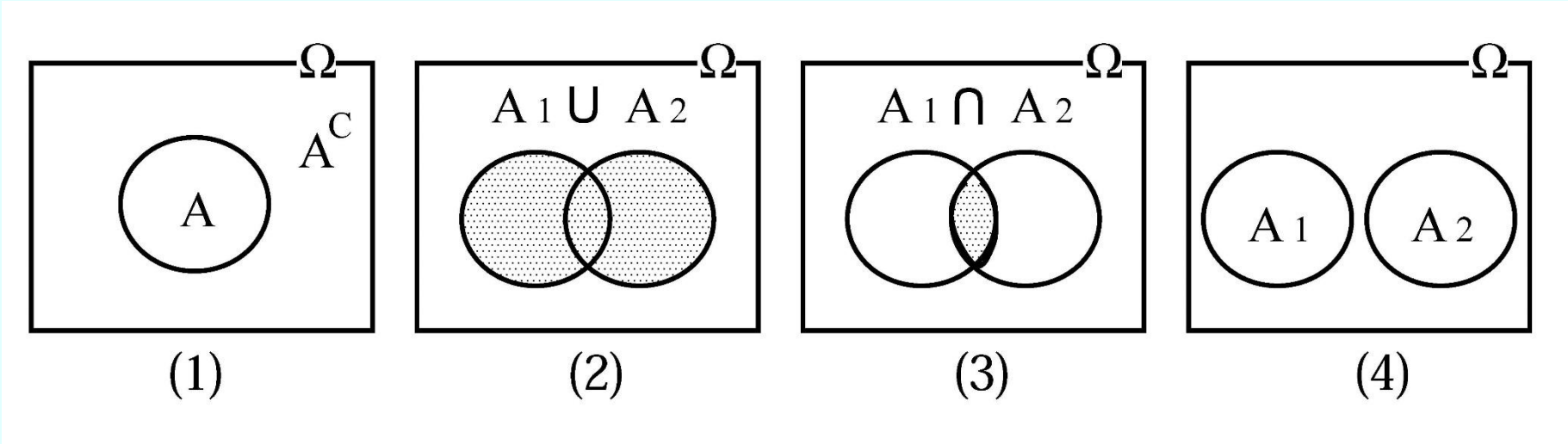
(E2) Sum Event (Union, OR):

$\boldsymbol{A}_1 \cup \boldsymbol{A}_2 = \{\omega \in \boldsymbol{\Omega}: \omega \in \boldsymbol{A}_1 \text{ or } \omega \in \boldsymbol{A}_2\}$ ("$\boldsymbol{A}_1$ or $\boldsymbol{A}_2$")

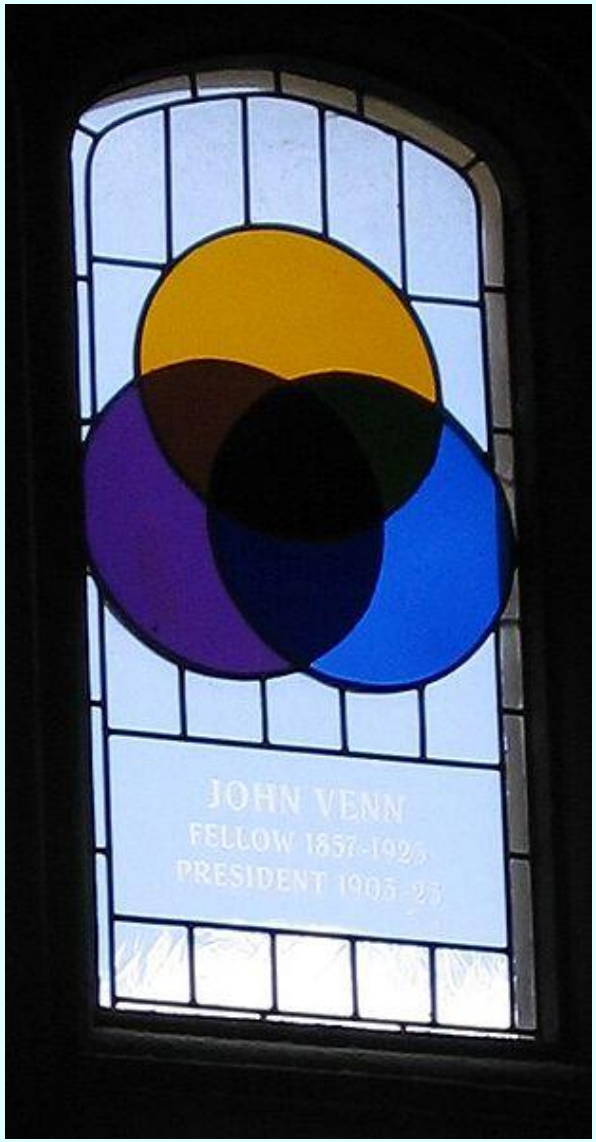
(E3) Product Event (Intersection, AND):

$\boldsymbol{A}_1 \cap \boldsymbol{A}_2 = \{\omega \in \boldsymbol{\Omega}: \omega \in \boldsymbol{A}_1 \text{ and } \omega \in \boldsymbol{A}_2\}$ ("$\boldsymbol{A}_1$ and $\boldsymbol{A}_2$")

(E4) Exclusive Event: $\boldsymbol{A}_1 \cap \boldsymbol{A}_2 = \boldsymbol{0}$



Draw Venn diagrams!



Borel set: $\boldsymbol{B}$

A Borel set $\boldsymbol{B}$ is any set in a topological space that can be formed from open sets (or, equivalently, from closed sets) through the operations of countable union, countable intersection, and relative complement.

(B1) $\boldsymbol{\Omega} \in \boldsymbol{B}$.

(B2) If $\boldsymbol{A} \in \boldsymbol{B}$ then $\boldsymbol{A}^C \in \boldsymbol{B}$.

(B3) If $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots \in \boldsymbol{B}$ then $\bigcup_{i=1}^{\infty} \boldsymbol{A}_i \; (= \boldsymbol{A}_1 \cup \boldsymbol{A}_2 \cup \cdots) \in \boldsymbol{B}$.

From (B1)-(B3),

(B4) $\mathbf{0} \in \boldsymbol{B}$.

(B5) If $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots \in \boldsymbol{B}$, then $\bigcap_{i=1}^{\infty} \boldsymbol{A}_i \; (= \boldsymbol{A}_1 \cap \boldsymbol{A}_2 \cap \cdots) \in \boldsymbol{B}$.

Borel set: Example

- Dice throwing

$$\boldsymbol{\Omega} = \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}, \mathbf{6}\}$$

$$\begin{aligned}
\boldsymbol{B} = \{&\emptyset, \{\mathbf{1}\}, \cdots, \{\mathbf{6}\}, \\
&\{\mathbf{1}, \mathbf{2}\}, \cdots \{\mathbf{5}, \mathbf{6}\}, \\
&\{\mathbf{1}, \mathbf{2}, \mathbf{3}\}, \cdots, \{\mathbf{4}, \mathbf{5}, \mathbf{6}\}, \\
&\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}, \{\mathbf{3}, \mathbf{4}, \mathbf{5}, \mathbf{6}\}, \\
&\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}\}, \cdots \{\mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}, \mathbf{6}\}, \\
&\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}, \mathbf{6}\}\}
\end{aligned}$$

Borel set = “a set of all possible events”

Probability

- Probability of event $\boldsymbol{A}$: $P(\boldsymbol{A})$
  mapping from event $\boldsymbol{A}$ to some numbers $P(\boldsymbol{A})$.
- **Properties of** $P(\boldsymbol{A})$ (Axioms of Probability theory)

  (P1) $0 \leq P(\boldsymbol{A}) \leq 1$

  (P2) $P(\boldsymbol{\Omega}) = 1$

  (P3) If $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots$ are exclusive events, then

  $P(\cup_{i=1}^{\infty} \boldsymbol{A}_i) = \sum_{i=1}^{\infty} P(\boldsymbol{A}_i)$ (Complete Additiveness)

- **Definition of Probability Space**:

  Sample space $\boldsymbol{\Omega}$, Borel set $\boldsymbol{B}$, and probability P define probability space.

Basic properties

(P4) $P(\mathbf{0}) = 0$

[Proof] Assuming $\boldsymbol{A}_i = \mathbf{0}$ $(i = 1, 2, \cdots)$, then

$$\cup_{i=1}^{\infty} \boldsymbol{A}_i = \cup_{i=1}^{\infty} \mathbf{0} = \mathbf{0}.$$

On the other hand, from the assumption,

$$\boldsymbol{A}_i \cap \boldsymbol{A}_j = \mathbf{0} \cap \mathbf{0} = \mathbf{0} \ (i \neq \mathrm{j}).$$

According to Axiom (P3),

$$P(\mathbf{0}) = P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{\infty} P(\mathbf{0}).$$

Since $\mathbf{0} \in \boldsymbol{\Omega}, P(\mathbf{0}) \geq 0$ (due to Axiom (P1)). The above equation holds only if $P(\mathbf{0}) = 0$.

(P5) If events $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots$ are **mutually exclusive**, then

$$P(\cup_{i=1}^{n} \boldsymbol{A}_i) = \sum_{i=1}^{n} P(\boldsymbol{A}_i).$$

[Proof] Setting $\boldsymbol{A}_{n+1} = \boldsymbol{A}_{n+2}, = \cdots = \boldsymbol{0}$,

$\boldsymbol{A}_1, \boldsymbol{A}_2, \cdots, \boldsymbol{A}_n, \boldsymbol{A}_{n+1}, \cdots$ are mutually exclusive.

Moreover, $\cup_{i=1}^{n} \boldsymbol{A}_i = \cup_{i=1}^{\infty} \boldsymbol{A}_i$.

According to Axiom (P3) and Property (P4),

$$P(\cup_{i=1}^{n} \boldsymbol{A}_i) = P(\cup_{i=1}^{\infty} \boldsymbol{A}_i)$$
$$= \sum_{i=1}^{n} P(\boldsymbol{A}_i) + \sum_{i=n+1}^{\infty} P(\boldsymbol{A}_i)$$
$$= \sum_{i=1}^{n} P(\boldsymbol{A}_i) + \sum_{i=n+1}^{\infty} P(\boldsymbol{0}) = \sum_{i=1}^{n} P(\boldsymbol{A}_i)$$

(P6) $P(\boldsymbol{A}^C) = 1 - P(\boldsymbol{A})$

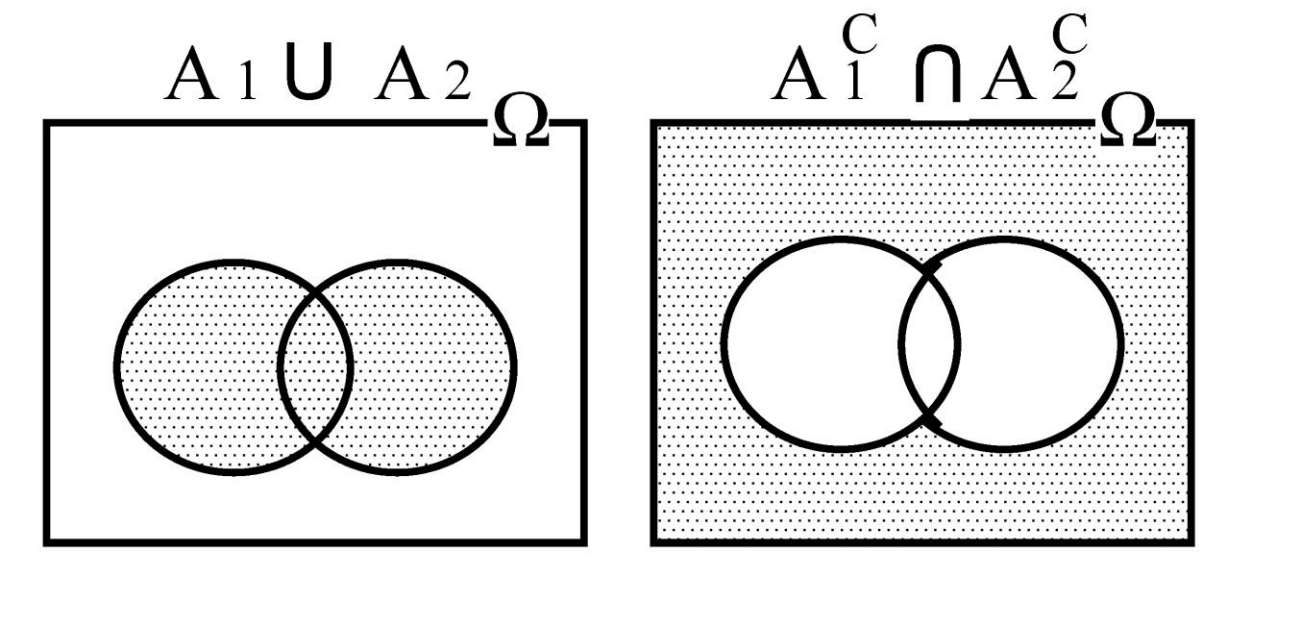
(P7) $P(\cup_i \boldsymbol{A}_i) = 1 - P(\cap_i \boldsymbol{A}_i^C)$

De Morgan's laws:

$$(\cap_i \boldsymbol{A}_i)^C = \cup_i \boldsymbol{A}_i^C, \ (\cup_i \boldsymbol{A}_i)^C = \cap_i \boldsymbol{A}_i^C$$

De Morgan's laws in case of 2 sets

$$(\boldsymbol{A}_1 \cup \boldsymbol{A}_2)^C = \boldsymbol{A}_1^C \cap \boldsymbol{A}_2^C$$



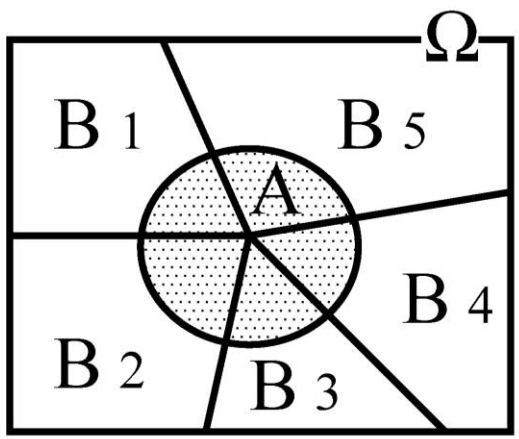
(P8) If sequence of mutually exclusive events, $\boldsymbol{B}_1, \boldsymbol{B}_2, \cdots$, is such that :

$\cup_i \boldsymbol{B}_i = \boldsymbol{\Omega}$, then $P(\boldsymbol{A}) = \sum_i \ P(\boldsymbol{A} \cap B_i)$.

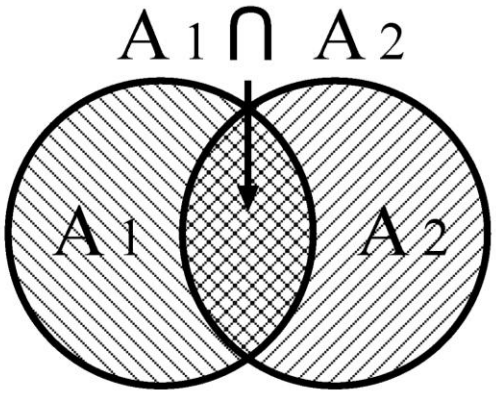
(P9) If $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are not exclusive, then

$P(\boldsymbol{A}_1 \cup \boldsymbol{A}_2) = P(\boldsymbol{A}_1) + P(\boldsymbol{A}_2) - P(\boldsymbol{A}_1 \cap \boldsymbol{A}_2)$.

(8)



(9)



(P10) For any $\boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{A}_3$,

$$\begin{aligned}&\mathrm{P}(\boldsymbol{A}_1 \cup \boldsymbol{A}_2 \cup \boldsymbol{A}_3)\\&= P(\boldsymbol{A}_1) + P(\boldsymbol{A}_2) + P(\boldsymbol{A}_3)\\&- \{P(\boldsymbol{A}_1 \cap \boldsymbol{A}_2) + P(\boldsymbol{A}_2 \cap \boldsymbol{A}_3) + P(\boldsymbol{A}_3 \cap \boldsymbol{A}_1)\}\\&\qquad + P(\boldsymbol{A}_1 \cap \boldsymbol{A}_2 \cap \boldsymbol{A}_3)\end{aligned}$$

(P11) General case: Denoting

$S_m = \sum_{i_1 \le i_2 \le \cdots \le i_m} P(\boldsymbol{A}_{i_1} \cap \boldsymbol{A}_{i_2} \cap \cdots \cap \boldsymbol{A}_{i_m})$, then

$P(\bigcup_{i=1}^{n} \boldsymbol{A}_i) = S_1 - S_2 + S_3 - S_4 + \cdots + (-1)^{n-1} S_n$.

Note: Additive terms on the right side are subject to all combination of S_n choosing *n* events from $1, 2, \cdots, n$.

The combination number is ${}_nC_m = \binom{n}{m} = \frac{n!}{(n-m)!m!}$.

[Exercise] Consider an experiment of drawing two cards at random from a bag containing four cards marked with the integers 1 through 4.

(a) Find the sample space S1 of the experiment if the first card is replaced before the second is drawn.

(b) Find the sample space S2 of the experiment if the first card is not replaced.

[Answer]

(a) The sample space S1 contains 16 ordered pairs (i, j), 1 ≤ i ≤ 4, 1 ≤ j ≤ 4, where the first number indicates the first number drawn. Thus,

$$S_1 = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) & (3,4) \\ (4,1) & (4,2) & (4,3) & (4,4) \end{Bmatrix}$$

[Answer]

(b) The sample space S2 contains 12 ordered pairs (i, j), i ≠ j, 1 ≤ i ≤ 4, 1 ≤ j ≤ 4, where the first number indicates the first number drawn. Thus,

$$S_2 = \begin{Bmatrix} (1,2) & (1,3) & (1,4) \\ (2,1) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,4) \\ (4,1) & (4,2) & (4,3) \end{Bmatrix}$$

[Exercise]

Given that P(A) = 0.9, P(B) = 0.8, and P(A ∩ B) = 0.75, find

(a) $P(A \cup B)$

(b) $P(A \cap \overline{B})$

(c) $P(\overline{A} \cap \overline{B})$.

[Answer]

(a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.9 + 0.8 - 0.75 = 0.95$$

(b) $P(A \cap \bar{B}) = P(A) - P(A \cap B)$

$$= 0.9 - 0.75 = 0.15$$

(c) By Morgan's Law:

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - 0.95 = 0.05$$

Continuity of Probability

(P12) Consider infinite sequence of events $\boldsymbol{A}_1, \boldsymbol{A}_2, \cdots$ such that $\boldsymbol{A}_1 \subset \boldsymbol{A}_2 \subset \cdots$. For $\boldsymbol{A} = \bigcup_{i=1}^{\infty} \boldsymbol{A}_i$,

$$P(\boldsymbol{A}) = \lim_{i \to \infty} P(\boldsymbol{A}_i)$$

(P13) Consider infinite sequence of events $\boldsymbol{A}_1, \boldsymbol{A}_2, \cdots$ such that $\boldsymbol{A}_1 \supset \boldsymbol{A}_2 \supset \cdots$. For $\boldsymbol{A} = \bigcap_{\boldsymbol{i=1}}^{\infty} \boldsymbol{A}_i$,

$$P(\boldsymbol{A}) = \lim_{i \to \infty} P(\boldsymbol{A}_i)$$

Conditional Probability

Conditional Probability

First event has an influence on probability of next event.

B: First event

A: Next event

Conditional probability $P(\boldsymbol{A}|\boldsymbol{B})$ of event ***A*** assuming event ***B*** is

$$P(\boldsymbol{A}|\boldsymbol{B}) = \frac{P(\boldsymbol{A} \cap \boldsymbol{B})}{P(\boldsymbol{B})}$$

[Exercise]

Prove that

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

[Answer]

$$P(A \cap B \cap C) = P(A)\frac{P(B \cap A)}{P(A)}\frac{P(C \cap A \cap B)}{P(A \cap B)}$$

$$= P(C \cap A \cap B) = P(A \cap B \cap C)$$

Properties of Conditional Probability

(C1) If we fix $\boldsymbol{B}$ and denote as $P(\boldsymbol{A}|\boldsymbol{B}) \equiv P^*(\boldsymbol{A})$,

(P1) - (P13) hold for $P^*(\boldsymbol{A})$.

(C2) $P(\boldsymbol{A} \cap \boldsymbol{B}) = P(\boldsymbol{A}|\boldsymbol{B})P(\boldsymbol{B})$,

(C3) If events $\boldsymbol{B}_1, \boldsymbol{B}_2, \cdots$ are mutually exclusive and $\bigcup_i \boldsymbol{B}_i = \boldsymbol{\Omega}$, then, for any event $\boldsymbol{A}$,

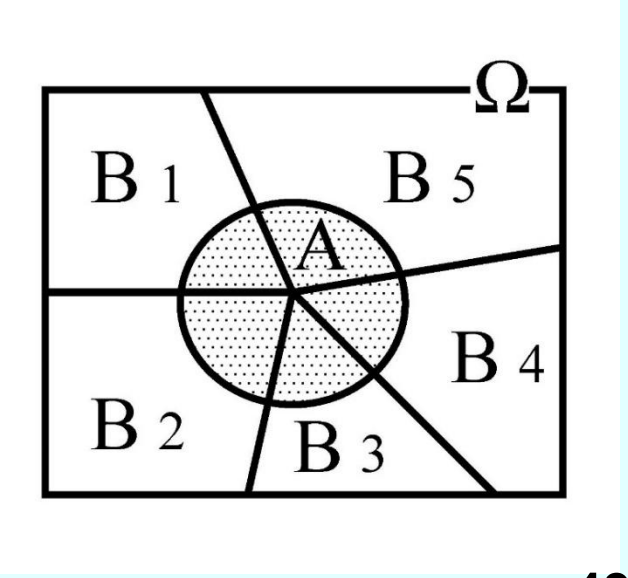
$P(\boldsymbol{A}) = \sum_i P(\boldsymbol{B}_i)P(\boldsymbol{A}|\boldsymbol{B}_i)$

(Total Probability Theorem)

(C4) If $P(\boldsymbol{A}) \geq 0$,

$$P(\boldsymbol{B}_i|\boldsymbol{A}) = \frac{P(\boldsymbol{A}|\boldsymbol{B}_i)P(\boldsymbol{B}_i)}{\sum_j P(\boldsymbol{A}|\boldsymbol{B}_j)P(\boldsymbol{B}_j)}$$

(Bayes' Theorem)



[Exercise] Company C buys the same product from three companies (B_1, B_2, B_3). The purchase ratio is $B_1 = 0.5$, $B_2 = 0.3$, and $B_3 = 0.2$. Probability that a new product is broken within one year is given for each company as $B_1 = 0.015$, $B_2 = 0.025$, and $B_3 = 0.035$.

- Suppose now that one product is broken within one year of purchase. We denote this event by *A*. What is the probability that the broken product was purchased from company B_1, B_2, or B_3.

[Answer] $P(B_1) = 0.5, P(B_2) = 0.3, P(B_3) = 0.2$

$P(A|B_1) = 0.015, P(A|B_2) = 0.025,$

$P(A|B_3) = 0.035$

From Total Probability Theorem (C3), $P(A) = \sum_{i=1}^{3} P(A|B_i)P(B_i) = 0.0075 + 0.0075 + 0.007$

$= 0.022$

Using Bayes' Theorem (C4)

$$P(B_1|A) = \frac{0.0075}{0.022} \approx 0.34, P(B_2|A) = \frac{0.0075}{0.022} \approx 0.34$$

$$P(B_3|A) = \frac{0.0070}{0.022} \approx 0.32$$

Independency

(1) If events ***A*** and ***B*** are **mutually independent**, then

$$P(\boldsymbol{A} \cap \boldsymbol{B}) = P(\boldsymbol{A})P(\boldsymbol{B}).$$

(2) Necessary and sufficient condition for mutual independence of *n* events, $\boldsymbol{A}_1, \boldsymbol{A}_2, \cdots, \boldsymbol{A}_n$, is that, for a set of arbitrarily chosen events, $\boldsymbol{A}_{i(1)}, \boldsymbol{A}_{i(2)}, \cdots, \boldsymbol{A}_{i(k)}$, the following holds

$$P\left(\boldsymbol{A}_{i(1)} \cap \boldsymbol{A}_{i(2)} \cap \cdots \cap \boldsymbol{A}_{i(k)}\right)$$
$$= P\left(\boldsymbol{A}_{i(1)}\right)P(\boldsymbol{A}_{i(2)}) \cdots P(\boldsymbol{A}_{i(k)})$$

[Exercise]

Show that if three events A, B, and C are independent, then A and $(B \cup C)$ are independent.

[Exercise]

At a college:

54% of students are female

25% of students are majoring in engineering.

15% of female students are majoring in engineering.

Event E = student is majoring in engineering

Event F = student is female

- Are events E and F independent? Use probabilities to justify your conclusion.
- Find $P(E \cap F)$

2. Random variable

- **Definition**:

  If real valued function $\boldsymbol{X}(\omega)$ $(\omega \in \boldsymbol{\Omega})$ defined on probability space satisfies $\{\omega: \boldsymbol{X}(\omega) \leq x\} \in \boldsymbol{B}$ for any real value x, $\boldsymbol{X}(\boldsymbol{\Omega})$ is called ***random variables***.

- **Example**

  From a box with many balls with different color, pick up one ball. One obtains a coupon card corresponding to the color of the ball.

Color (Even) ω	White	Green	Yellow	Blue	Red
Coupon (Random variable) $X(\boldsymbol{\Omega})$	500	1000	2000	4000	6000
Probability	0.72	0.15	0.1	0.02	0.01

P(white)=$\{\omega: X(\boldsymbol{\Omega}) \leq 500\}$

3. Distribution function

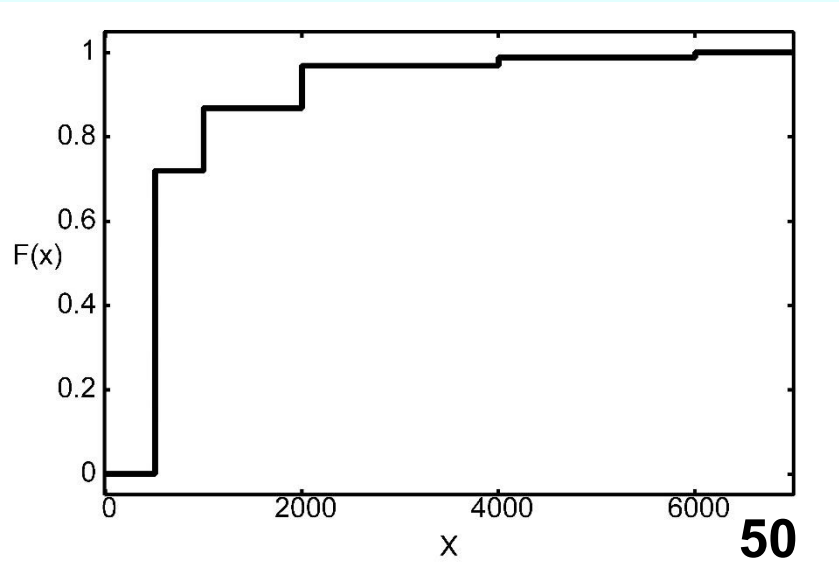
- **Definition**:

$$F_{\boldsymbol{X}}(x) = P(\{\omega: \boldsymbol{X}(\omega) \leq x\}), -\infty \leq x \leq \infty$$

Left-hand-side can be simply described as *F(x)*
Right-hand-side as $P(\boldsymbol{X} \leq x)$.

- **Example**

$$\begin{aligned} &F(2000) \\ &\quad = P(\{\omega: X(\omega) \leq 2000\}) \\ &\quad = P(\{\text{Yellow} \cup \text{Green} \cup \text{White}\}) \\ &\quad = 0.72 + 0.15 + 0.1 \\ &\quad = 0.97 \end{aligned}$$



Properties of the distribution function

(PD1) Non-decreasing

If $a < b$,

then $F(a) \leq F(b)$.

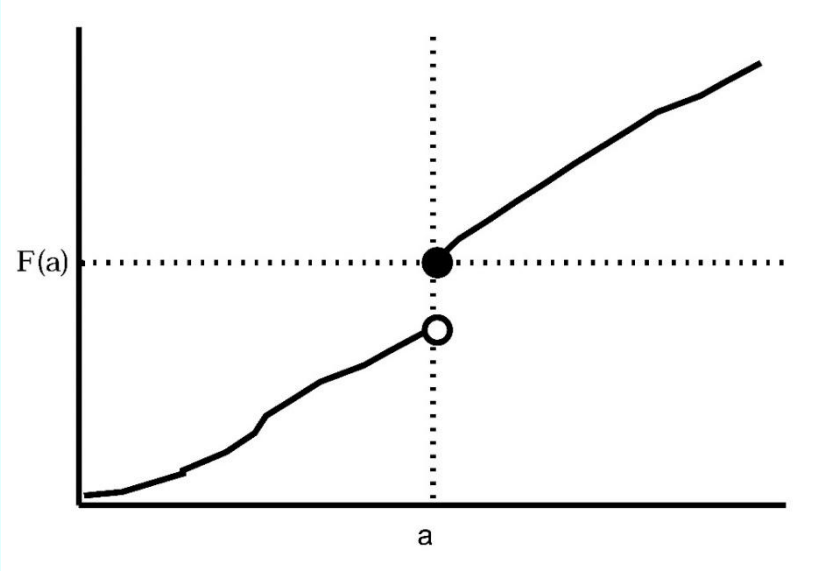
(PD2) Continuous from the right

$$\lim_{x \to a+0} F(x) = F(a)$$

(PD3) Limit

$$\lim_{x \to -\infty} F(x) = 0$$

$$\lim_{x \to \infty} F(x) = 1$$

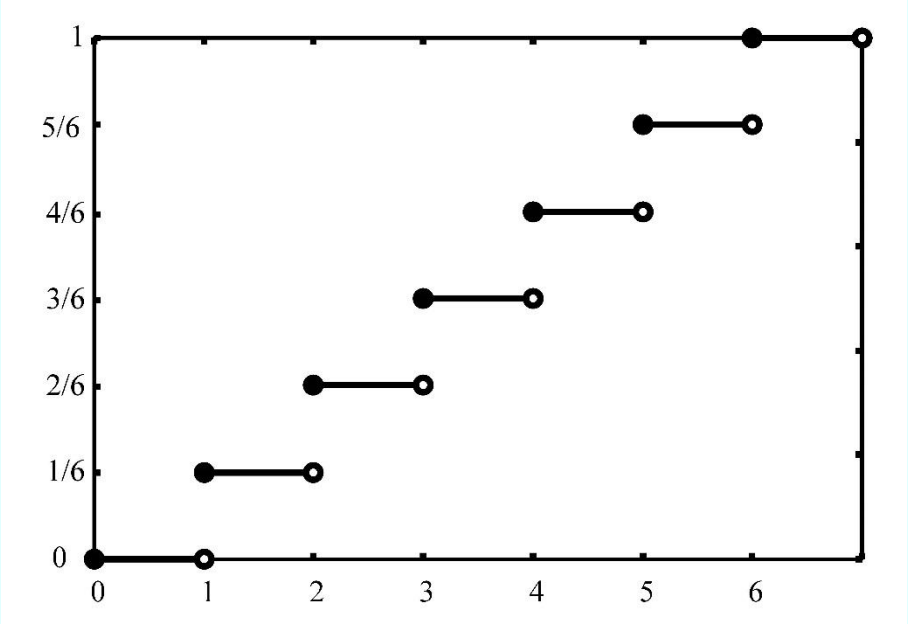


[Example]

- Trial $\boldsymbol{S}$: Throw dice
- Sample Point ω: 1, 2, 3, …
- Sample Space $\boldsymbol{\Omega}$: {1, 2, 3, 4, 5, 6}

- $F(\{1 \leq \omega < 2\}) = \frac{1}{6}$
- $F(2) = \frac{1}{3}$
- $F(3) = \frac{1}{2}$

  …
- $F(6) = 1$



[Exercise]

Consider the experiment of tossing a coin three times. Let X be the random variables giving the number of heads obtained. We assume that the tosses are independent and the probability of a head is p.

(a) What is the range of X?

(b) Find the probabilities P(X = 0), P(X = 1), P(X = 2), and P(X = 3).

[Answer]

(a) Since we are tossing the coin three times, the number of heads we can get can be 0, 1, 2, or 3. Therefore, the range of X is {0, 1, 2, 3}.

(b) If P(H) = p, then P(T) = 1 − p. Since the tosses are independent, we have

$$P(X = 0) = P[\{TTT\}] = (1 - p)^3$$

$$P(X = 1) = P[\{HTT\}] + P[\{THT\}] + P[\{TTH\}] = 3(1 - p)^2 p$$

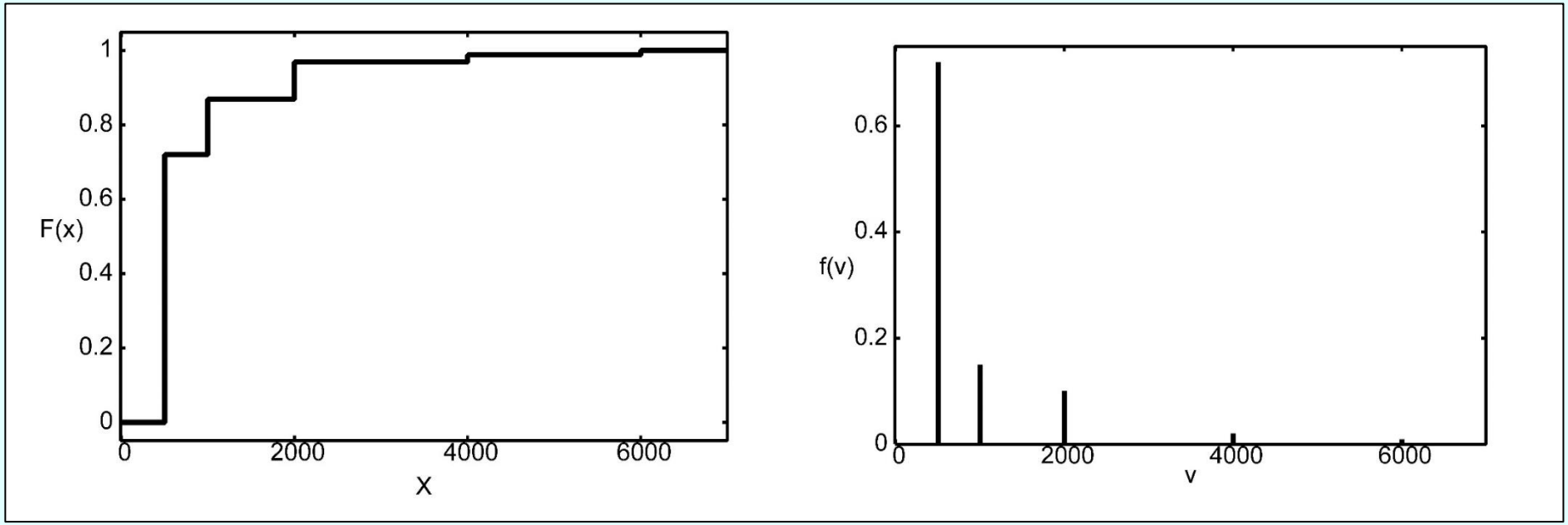
$$P(X = 2) = P[\{HHT\}] + P[\{HTH\}] + P[\{THH\}] = 3p^2(1 - p)$$

$$P(X = 3) = P[\{HHH\}] = p^3$$

4. Discrete-type Distribution Function

- Possible values $\boldsymbol{V} = \{v_0, v_1, v_2, \cdots\}$ that $\boldsymbol{X}$ can take is finite or countable. Taking $\varepsilon > 0$ in a way that no elements except v_k are included in the interval $[v_k - \varepsilon, v_k]$, the **probability mass function (PMF)** (probability that $\boldsymbol{X}$ takes a value of):

$$f(v_k) = P(X = v_k) \text{ is } f(v_k) = F(v_k) - F(v_k - \varepsilon).$$



- **Properties**

  (DT1) $F(x) = \sum_{v_k \le x} f(v_k)$

  (DT2) For all *k*, $f(v_k) \ge 0$

  (DT3) $\sum_k f(v_k) = 1$

- **Example of discrete-type distribution function**

  (Ex1) **Bernoulli distribution**: $B(1; p), 0 < p < 1$

  Random variable that can take only two values (i.e., biased coin)

$$V = \{0, 1\}$$
$$f(0) = 1 - p$$
$$f(1) = p$$

(Ex2) **Binominal distribution**: $B(n;p), 0 < p < 1$

Consider a lottery that has *p* percent of winning tickets. Try the lottery for *n* times. Probability to draw the winning ticket for k times yields binominal distribution. Probability to win *k* times and loose $(n-k)$ times is $p^k = (1-p)^{n-k}$. Combination of choosing *k* objects among *n* is ${}_nC_k = \binom{n}{k}$.

Hence, $f(k) = \binom{n}{k} p^k (1-p)^{n-k}$

(Ex3) **Poisson distribution**: $P_o(\lambda)$

Binominal distribution in the limit of $n \to \infty,\ p \to 0$,

$$np = \lambda,\ f(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

Stirling's formula $n! \approx \sqrt{2\pi n}\ n^n e^{-n}$

[Exercise]

I roll two dice and observe two numbers X and Y.

a) Find sample space of X and Y, and the PMFs of X and Y

b) Find $P(X = 2, Y = 6)$

c) Find $P(X > 3 | Y = 2)$

[Answer]

a) **Sample Space of X and Y**:

Since each die has 6 sides, the sample space for both X and Y is the same:

S = {1, 2, 3, 4, 5, 6}

PMFs of X and Y:

Since the dice are fair, every outcome has an equal "mass."

$$P(Y = y) = \frac{1}{6} \quad \text{for } y \in \{1,2,3,4,5,6\} P(X = x) = \frac{1}{6} \quad \text{for } x \in \{1,2,3,4,5,6\}$$

[Answer]

b) Find $P(X = 2, Y = 6)$

Because the rolls are independent, we multiply their individual probabilities:

$$P(X = 2, Y = 6) = P(X = 2) \times P(Y = 6)$$

$$P(X = 2, Y = 6) = \frac{1}{6} \times \frac{1}{6} = \mathbf{\frac{1}{36}}$$

[Answer]

c) Find $P(X > 3|Y = 2)$

Since the outcome of the second die (Y=2) has no effect on the first die (X), these events are independent. Knowing what Y is doesn't change the odds for X.

Therefore:

$$P(X > 3 \mid Y = 2) = P(X > 3)$$

$$P(X > 3) = \frac{3}{6} = \mathbf{\frac{1}{2}} \text{ (or 0.5)}$$

5. Continuous-type Distribution Function

- **Distribution function:**

  $F(x)$ can be described as

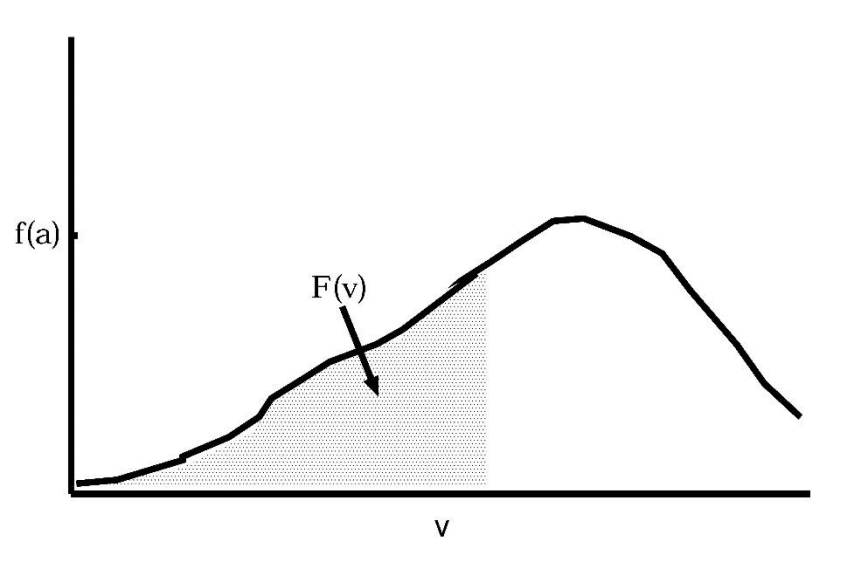
$$F(x) = \int_{-\infty}^{x} f(v)dv$$

- **Probability density** is

$$f(x) = \frac{d}{dx}F(x)$$

(CT1) $f(x) > 0, -\infty < x < \infty$

(CT2) $\int_{-\infty}^{\infty} f(x)dx = 1$



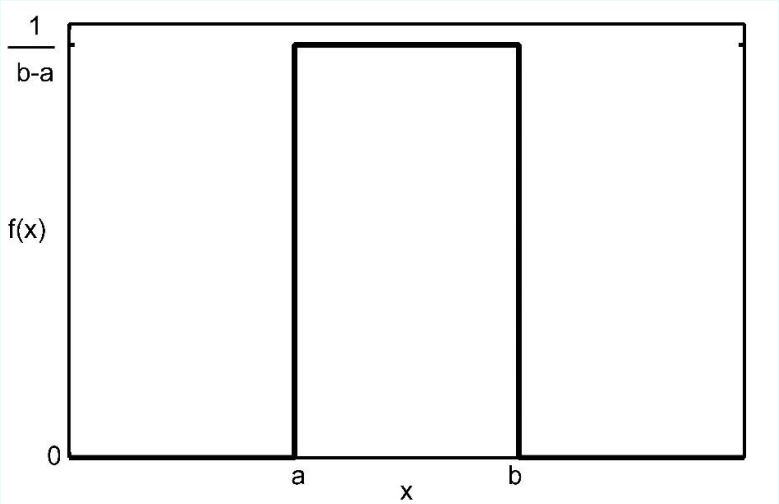
Example of continuous-type distribution function

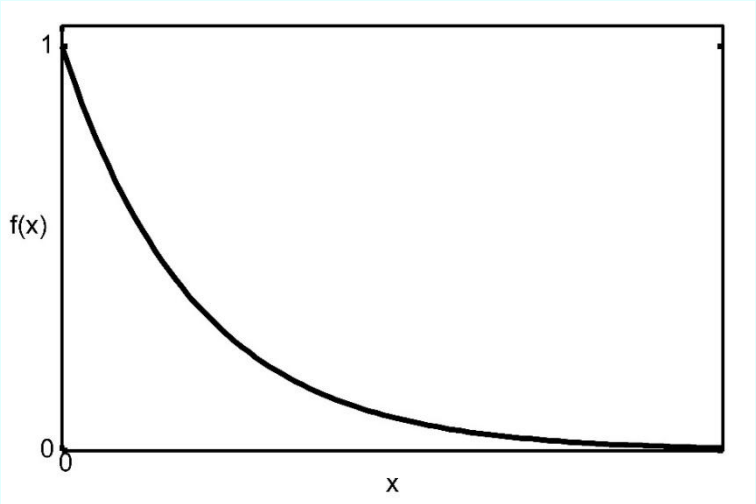
(Ex1) **Uniform distribution**: $U(a,b), a < b$

$$U(a,b) = f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

(Ex2) **Exponential distribution**: $E_x(\alpha), \alpha > 0$

$$E_x(\alpha) = f(x) = \begin{cases} \alpha e^{-\alpha x} & x \ge 0 \\ 0, & x < 0 \end{cases}$$





(Ex3) **Gamma distribution**: $G(\alpha, \nu)$

$$G(\alpha, \nu) = f(x) = \begin{cases} \frac{1}{\Gamma(\nu)} \alpha^{\nu} x^{\nu-1} e^{-\alpha x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

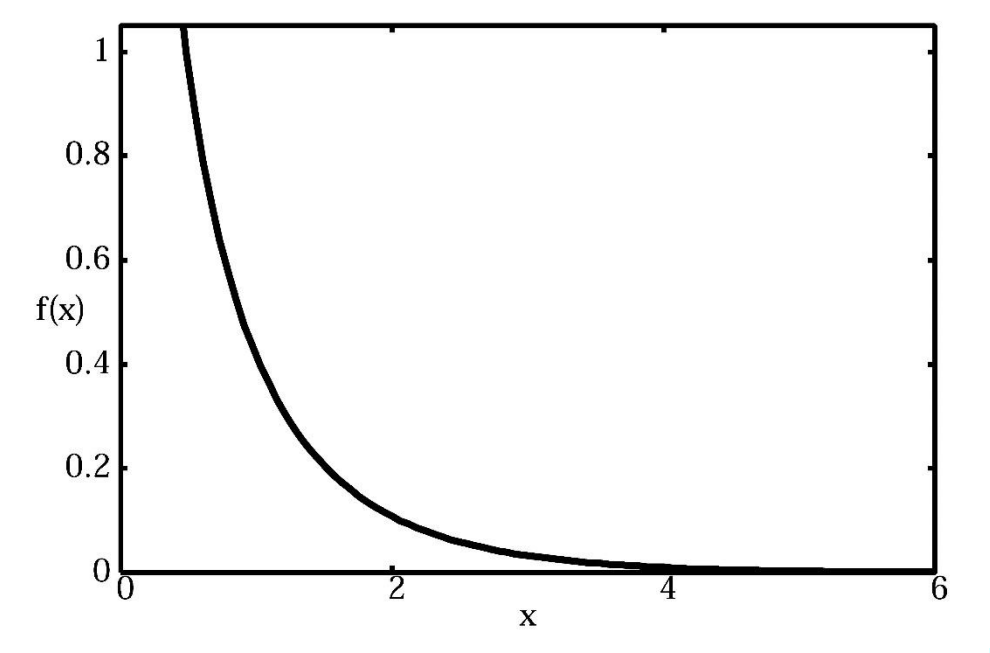
where $\Gamma(\nu) = \int_0^{\infty} t^{\nu-1} e^{-t} dt$.

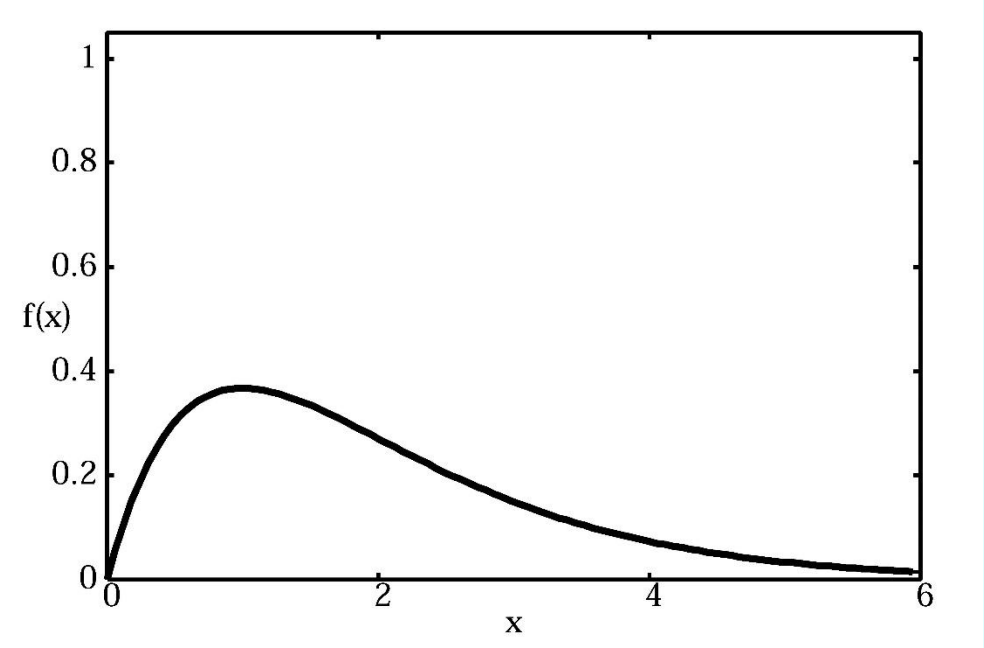
- Properties of Γ :
  - $\Gamma(\nu) = (\nu - 1)\Gamma(\nu - 1),\ \Gamma(1) = 1$
  - $\Gamma(1/2) = \sqrt{\pi}$

  Especially, if *n* is an integer,
  - $\Gamma(n) = (n - 1)!$
  - $\Gamma\left(n + \frac{1}{2}\right) = \left(n - \frac{1}{2}\right)\left(n - \frac{3}{2}\right)\left(n - \frac{5}{2}\right) \cdots \frac{1}{2}\sqrt{\pi}$

When $\nu = 1$, equivalent to $E_x(\alpha)$.

- $G\left(\frac{1}{2}, \frac{n}{2}\right)$: χ^2 (chi-square) distribution with *n* degree of freedom (often used for ANOVA in statistics).



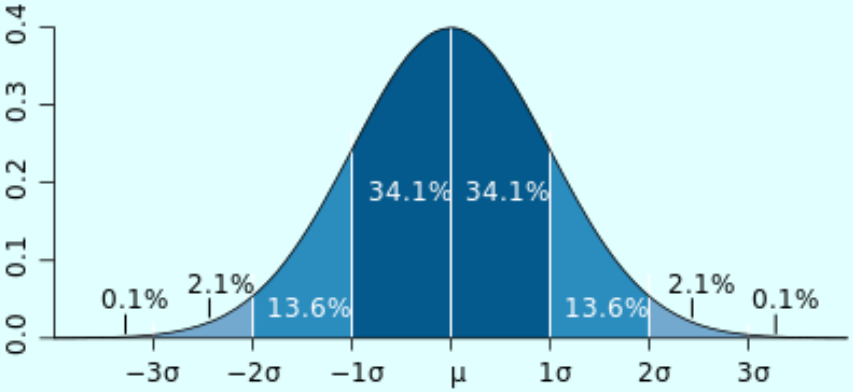


(Ex4) **Gaussian distribution**:

$$N(\mu, \sigma^2), -\infty < \mu < \infty, \sigma > 0$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], -\infty < x < \infty$$

- In case of $N(0,1)$, **standard normal distribution**
- Probability density: $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}x^2\right]$
- Distribution function: $\boldsymbol{\Phi}(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}t^2\right] dt$
- Properties of $\phi(x)$
  - $\phi(x) = \phi(-x)$
  - $\boldsymbol{\Phi}(x) = 1 - \boldsymbol{\Phi}(-x)$
  - $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$



[Exercise] Suppose $\boldsymbol{X}$ is a random variable of $N(0,1)$. With respect to $\boldsymbol{Y} = \boldsymbol{X}^2$, find its distribution function $F_{\boldsymbol{Y}}(y)$ and the probability density $f_{\boldsymbol{Y}}(y)$.

[Answer] If $y < 0, F_{\boldsymbol{Y}}(y) = 0,\ f_{\boldsymbol{Y}}(y) = 0.$

If $y \geq 0$,

$$F_{\boldsymbol{Y}}(y) = P(\boldsymbol{Y} \leq y) = P(\boldsymbol{X}^2 \leq y) = P(-\sqrt{y} \leq \boldsymbol{X} \leq \sqrt{y})$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \phi(x)dx = 2\int_0^{\sqrt{y}} \phi(x)dx,$$

$$f_{\boldsymbol{Y}}(y) = \frac{dF_{\boldsymbol{Y}}(y)}{dy} = \frac{d}{dy}\left\{2\int_0^{\sqrt{y}} \phi(x)dx\right\} = 2\phi\left(\sqrt{y}\right)\frac{1}{2\sqrt{y}}$$

$$= \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} \exp\left(-\frac{1}{2}y\right) = G\left(\frac{1}{2}, \frac{1}{2}\right)$$

This is χ^2 distribution with $n = 1$ degrees of freedom.

6. Joint Distribution

- Consider more than n random variables $\boldsymbol{X}_1(\omega), \boldsymbol{X}_2(\omega), \cdots, \boldsymbol{X}_n(\omega)$ simultaneously, and examine their interrelation.
- With regard to event $\{\omega: \boldsymbol{X}_1(\omega) \leq x_1, \boldsymbol{X}_2\ (\omega) \leq x_2, \cdots, \boldsymbol{X}_n(\omega) \leq x_n\} = \bigcap_{k=1}^{n}\{\omega: \boldsymbol{X}_k(\omega) \leq x_k\}$

  the distribution function is defined as

$$F(x_1, x_2, \cdots, x_n) = P(\bigcap_{k=1}^{n}\{\omega:\ \boldsymbol{X}_k(\omega) \leq x_k\})$$

Case of $n = 2$: $F(x_1, x_2) = \int_{-\infty}^{x_1}\int_{-\infty}^{x_2} f(u, v)dudv$

- Marginal distribution function of $\boldsymbol{X}_1$: $F_1(x) = F(x, +\infty)$
- Marginal distribution function of $\boldsymbol{X}_1$: $F_2(x) = F(+\infty, x)$
- Conditional distribution function given

$$\boldsymbol{X}_2 < c: F(x, c)/F(+\infty, c)$$

[Example] Two-Dimensional Gaussian Distribution

Density function

$$f(u,v) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}Q(u,v)\right\}$$

where

$$Q(u,v) = \frac{1}{1-\rho^2}\left\{\left(\frac{u-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{u-\mu_1}{\sigma_1}\right)\left(\frac{v-\mu_2}{\sigma_2}\right) + \left(\frac{v-\mu_2}{\sigma_2}\right)^2\right\}$$

$$\rho = \frac{C}{\sigma_1\sigma_2},$$

$$C = E\{(u-\mu_1)(v-\mu_2)\}$$ (**Covariance**)

7. Independency of random variables

Consider *n* random variables $\boldsymbol{X_1}(\omega), \boldsymbol{X}_2(\omega), \quad \cdots, \boldsymbol{X}_n(\omega)$.
If their joint distribution function $F(x_1, x_2, \cdots, x_n)$ is equal to the product of distribution functions of each random variable

$$F(x_1, x_2, \cdots, x_n) = F(x_1)F(x_2)\cdots F(x_n).$$

- *n* random variables are mutually independent.

- The same holds for probability density as

$$f(x_1, x_2, \cdots, x_n) = f(x_1)f(x_2)\cdots f(x_n)$$

Case of 2-dimentional normal distribution function.
Suppose $\rho = 0$ (covariance is zero).

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{-\frac{1}{2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}$$
$$= f_1(x_1) f_2(x_2)$$

where

$$f_i(x_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left\{-\frac{1}{2}\left[\left(\frac{x_i-\mu_i}{\sigma_i}\right)^2\right]\right\} \quad (i = 1, 2)$$

8. Mean

- **Discrete-type Distribution Function**:

  When probability function $f(v_k)$ is given by

  $$f(v_k) = P(\boldsymbol{X} = v_k),$$

  $$E(\boldsymbol{X}) = \sum_k v_k f(v_k)$$

  is called ***mean*** or ***expected value***.

- **Remark**

  Strictly speaking, the mean exists when

  $$\sum_k |v_k| f(v_k) < \infty$$

- **Example Dice**

  $$1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = 3.5$$

Continuous-type Distribution Function:

When the probability function of $\boldsymbol{X}$ is given by $f(x)$,

$$E(\boldsymbol{X}) = \int_{-\infty}^{\infty} x f(x) dx$$

Remark

Strictly speaking, the mean exists when

$$\int_{-\infty}^{\infty} |x| f(x) dx < \infty$$

Example

(Ex1) **Bernoulli distribution** $B(1; p)$

$E(\boldsymbol{X}) = 0 \times f(0) + 1 \times f(1) = 0 \times 1(1-p) + 1 \times p = p$

[Exercise]

Let $X \sim Uniform(a, b)$. Find $E(\boldsymbol{X})$.

[Answer]

$E(\boldsymbol{X}) = \frac{(a+b)}{2}$

(Ex2) **Binomial distribution** $B(n;p)$

$$
\begin{aligned}
E(\boldsymbol{X}) &= \sum_{k=0}^{n} k f(k) = \sum_{k=1}^{n} k f(k) \\
&= \sum_{k=1}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=1}^{n} n \binom{n-1}{k-1} p^k (1-p)^{n-k} \\
&= np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{(n-1)-(k-1)} \\
&= np \sum_{l=0}^{m} \binom{m}{l} p^l (1-p)^{m-l} \\
&= np
\end{aligned}
$$

(Ex3) **Poisson distribution** $P_0(\lambda)$

$$E(\boldsymbol{X}) = \sum_{k=0}^{\infty} kf(k) = \sum_{k=1}^{\infty} kf(k)$$

$$= \sum_{k=1}^{\infty} ke^{-\lambda}\frac{\lambda^k}{k!} = \lambda\sum_{k=1}^{\infty} e^{-\lambda}\frac{\lambda^{k-1}}{(k-1)!}$$

$$\{\text{substituting } m = k-1\}$$

$$= \lambda\sum_{m=0}^{\infty} e^{-\lambda}\frac{\lambda^m}{m!}$$

$$\left\{\text{from } \sum_{m=0}^{\infty} e^{-\lambda}\frac{\lambda^m}{m!} = \sum_{m=0}^{\infty} f(m) = 1\right\}$$

$$= \lambda$$

(Ex4) **Uniform distribution** $U(a,b)$

$$E(\boldsymbol{X}) = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

(Ex5) **Exponential distribution** $E_x(\alpha)$

$$E(\boldsymbol{X}) = \int_0^\infty x\alpha e^{-\alpha x} dx$$

$$= \left[x\alpha \frac{1}{-\alpha} e^{-\alpha x}\right]_0^\infty - \int_0^\infty \alpha \frac{1}{-\alpha} e^{-\alpha x} dx$$

$$= \int_0^\infty e^{-\alpha x} dx = \left[\frac{1}{-\alpha} e^{-\alpha x}\right]_0^\infty$$

$$= \frac{1}{\alpha}$$

(Ex6) **Gamma distribution** $G(\alpha, \nu)$

$$E(\boldsymbol{X}) = \int_0^\infty x \frac{1}{\Gamma(\nu)} \alpha^\nu x^{\nu-1} e^{-\alpha x} dx = \int_0^\infty \frac{1}{\Gamma(\nu)} t^\nu e^{-t} \frac{dt}{\alpha}$$

$$= \frac{1}{\alpha\Gamma(\nu)} \int_0^\infty t^\nu e^{-t} dt = \frac{\Gamma(\nu+1)}{\alpha\Gamma(\nu)} = \frac{\nu}{\alpha}$$

(Ex7) **Normal distribution** $N(\mu, \sigma^2)$

$$E(\boldsymbol{X}) = \int_{-\infty}^\infty x \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

$$= \int_{-\infty}^\infty (\mu + \sigma t) \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} t^2\right] dt$$

$$= \mu \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} t^2\right] dt$$

$$+ \int_{-\infty}^\infty \frac{\sigma t}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} t^2\right] dt = \mu$$

Properties of the Mean Value

(M1) For any real-value function $g(x)$, $g(\boldsymbol{X})$ is also a random variable. Mean value for $g(\boldsymbol{X})$ is given as follows.

- Case of discrete-type random variable

$$E\big(g(\boldsymbol{X})\big) = \sum_k g(v_k) f(v_k)$$

- Case of continuous-type random variable

$$E\big(g(\boldsymbol{X})\big) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

(M2) Conditional expectation

- Case of discrete-type random variable

$$f(v|y) = P(\boldsymbol{X} = v|\boldsymbol{Y} = y) \frac{P(\boldsymbol{X}=v, \boldsymbol{Y}=y)}{P(\boldsymbol{Y}=y)}$$

yields $$E(\boldsymbol{X}|\boldsymbol{Y} = y) = \sum_k v_k f(v_k|y)$$

$$E(\boldsymbol{X}) = \sum_y E(\boldsymbol{X}|\boldsymbol{Y} = y) P(\boldsymbol{Y} = y)$$

- Case of continuous-type random variable

$$f(v|y) = \frac{f(x,y)}{f(y)}$$

yields $$E(\boldsymbol{X}|\boldsymbol{Y} = y) = \int_{-\infty}^{\infty} x f(x|y)\, dx$$

$$E(\boldsymbol{X}) = \int_{-\infty}^{\infty} E(\boldsymbol{X}|\boldsymbol{Y} = y) f(y)\, dy$$

(M3) To unify the framework of computing the expected value for discrete-type and continuous-type random variables,

$$E(\boldsymbol{X}) = \int_{-\infty}^{\infty} x dF(x)$$

(**Reimann-Stieltjes integral**)

(a) $E(a\boldsymbol{X}+b)=aE(\boldsymbol{X})+b$

(b) When joint distribution function of $\boldsymbol{X}_1,\boldsymbol{X}_2,\cdots,\boldsymbol{X}_n$ is given by $F(x_1,x_2,\cdots,x_n)$,

$$E\big(g(\boldsymbol{X}_1,\boldsymbol{X}_2,\cdots,\boldsymbol{X}_n)\big)=$$

$$\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty}g(x_1,x_2,\cdots,x_n)dF(x_1,x_2,\cdots,x_n)$$

(c) $E(\boldsymbol{X}_1+\boldsymbol{X}_2+\cdots+\boldsymbol{X}_n)=E(\boldsymbol{X}_1)+E(\boldsymbol{X}_2)+\cdots+E(\boldsymbol{X}_n)$

(d) If $\boldsymbol{X}_1,\boldsymbol{X}_2,\cdots,\boldsymbol{X}_n$ are mutually independent,

$$E(\boldsymbol{X}_1\boldsymbol{X}_2\cdots\boldsymbol{X}_n)=E(\boldsymbol{X}_1)E(\boldsymbol{X}_2)\cdots E(\boldsymbol{X}_n)$$

This can be extended to

$$E(g_1(\boldsymbol{X}_1)g_2(\boldsymbol{X}_2)\cdots g_n(\boldsymbol{X}_n))=$$

$$E(g_1(\boldsymbol{X}_1))E(g_2(\boldsymbol{X}_2))\cdots E(g_n(\boldsymbol{X}_n))$$

9. Variance

Definition

Suppose mean of random variable $\boldsymbol{X}$ is given by $E(\boldsymbol{X}) = \mu$, $(\boldsymbol{X} - \mu)^2$ gives also a random variable. Mean of $(\boldsymbol{X} - \mu)^2$ is called ***variance*** of $\boldsymbol{X}$, which is denoted as $Var(\boldsymbol{X})$.

$$Var(\boldsymbol{X}) = E\{(\boldsymbol{X} - \mu)^2\} = \int_{-\infty}^{\infty} (x - \mu)^2 dF(x)$$
$$= \sum_k (v_k - \mu)^2 f(v_k)$$
$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Remark

$$\sum_k |v_k|^2 f(v_k) < \infty, \int_{-\infty}^{\infty} |x|^2 f(x) dx < \infty$$

Standard deviation

$$\sqrt{Var(\boldsymbol{X})}$$

Properties of variance

(V1) $Var(\boldsymbol{X}) \geq 0$

(V2) $Var(\boldsymbol{X}) = E(\boldsymbol{X}^2) - \{E(\boldsymbol{X})\}^2$

[Proof] $Var(\boldsymbol{X}) = E(\boldsymbol{X}^2 - 2\mu\boldsymbol{X} + \mu^2)$

$$= E(\boldsymbol{X}^2) - 2E(\mu\boldsymbol{X}) + E(\mu^2)$$

$$= E(\boldsymbol{X}^2) - 2\mu E(\boldsymbol{X}) + \mu^2$$

$$= E(\boldsymbol{X}^2) - 2\mu\mu + \mu^2 = E(\boldsymbol{X}^2) - \mu^2$$

(V3) $Var(a\boldsymbol{X} + b) = a^2 Var(\boldsymbol{X})$

[Proof] $Var(a\boldsymbol{X} + b) = E\left\{\left((a\boldsymbol{X} + b) - (a\mu + b)\right)^2\right\}$

$$= E\{a^2(\boldsymbol{X} - \mu)^2\}$$

$$= a^2 E\{(\boldsymbol{X} - \mu)^2\}$$

$$= a^2 Var(\boldsymbol{X})$$

Properties of variance

(V4) $Var(\boldsymbol{X}_1 + \boldsymbol{X}_2) = Var(\boldsymbol{X}_1) + Var(\boldsymbol{X}_2) + 2Cov(\boldsymbol{X}_1, \boldsymbol{X}_2)$

Covariance: $Cov(\boldsymbol{X}_1, \boldsymbol{X}_2) = E\{(\boldsymbol{X}_1 - \mu_1)(\boldsymbol{X}_2 - \mu_2)\}$

Correlation coefficient: $r(\boldsymbol{X}_1, \boldsymbol{X}_2) = \frac{Cov(\boldsymbol{X}_1, \boldsymbol{X}_2)}{\sqrt{Var(\boldsymbol{X}_1)Var(\boldsymbol{X}_2)}}$

If $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are mutually independent, $Cor(\boldsymbol{X}_1, \boldsymbol{X}_2) = 0$.

$\mathrm{r}(\boldsymbol{X}_1, \boldsymbol{X}_2) = 0$: No correlation

(V5) $Var(\boldsymbol{X}_1 + \boldsymbol{X}_2 + \cdots \boldsymbol{X}_n) =$

$$\sum_{i=1}^{n} Var(\boldsymbol{X}_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} Cor(\boldsymbol{X}_i, \boldsymbol{X}_j)$$

In particular, if $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ are mutually independent,

$$Var(\boldsymbol{X}_1 + \boldsymbol{X}_2 + \cdots + \boldsymbol{X}_n) = \sum_{i=1}^{n} Var(\boldsymbol{X}_i)$$

[Exercise]

Let X be a continuous random variable with PDF

$$f_x(x) = \begin{cases} x^2(2x + \frac{3}{2}), & 0 < x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

If $\mathrm{Y} = \frac{2}{X} + 3$, Find $Var(Y)$.

[Answer]

$$Var(Y) = Var\left(\frac{2}{X} + 3\right) = 4Var\left(\frac{1}{X}\right) = \frac{71}{36}$$

10. Moment

Definition

- *m*-th central moment (moment about mean)

$$\mu_m = E\{(\boldsymbol{X} - \mu)^m\}$$

- *m*-th moment about zero

$$\mu'_m = E\{\boldsymbol{X}^m\}$$

Remark

- If there exists an *m*-th moment, there exists all the moments lower than *m*-th order.
- If there exists no *m*-th moment, there exists no moment higher than *m*-th order.

Table of distribution functions

Name of the distribution and range of the parameters	Probability density function	Mean and Variance	Characteristic function
Bernoulli distribution $B(1;p)$ $(0 < p < 1, q = 1 - p)$	$p^k q^{1-k}$ $k = 0, 1$	p, pq	$pe^{jt} + q$
Binominal distribution $B(n;p)$ (n: Integer, $0 < p < 1, q = 1 - p$)	$\binom{n}{k} p^k q^{n-k}$ $k = 0, 1, 2, \cdots, n$	np, npq	$(pe^{jt} + q)^n$
Poisson distribution $Po(\lambda)$ $(\lambda > 0)$	$e^{-\lambda} \frac{\lambda^k}{k!}$ $k = 1, 2, \cdots$	λ, λ	$\exp[\lambda(e^{jt} - 1]$
Uniform distribution $U(a,b)$ $(-\infty < a < b < \infty)$	$\frac{1}{b-a}$ $a \leq x \leq b$	$(a+b)/2$ $(b-a)^2/12$	$\frac{e^{jbt} - e^{jat}}{j(b-a)t}$
Exponential distribution $Ex(\alpha)$ $(\alpha > 0)$	$\alpha e^{-\alpha x}$ $x \geq 0$	$1/\alpha, 1/\alpha^2$	$(1 - \frac{jt}{\alpha})^{-1}$
Gamma distribution $G(\alpha, \nu)$ $(\alpha, \nu > 0)$	$\frac{1}{\Gamma(\nu)} \alpha^\nu x^{\nu-1} e^{-\alpha x}$ $x \geq 0$	ν/α ν/α^2	$(1 - \frac{jt}{\alpha})^{-\nu}$
Normal distribution $N(\mu, \sigma^2)$ $(-\infty < \mu < \infty, \sigma > 0)$	$\frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$ $-\infty < x < \infty$	μ σ^2	$\exp[j\mu t - \frac{\sigma^2}{2} t^2]$

11. Characteristic Function

- **Definition** (Characteristic Function)

$$\varphi(t) \equiv E\left(e^{it\boldsymbol{X}}\right) = \int_{-\infty}^{\infty} e^{itx} dF(\boldsymbol{X})$$

- **Remark**

$$\begin{aligned} |\varphi(t)| &= \left|\int_{-\infty}^{\infty} e^{itx} dF(\boldsymbol{X})\right| \\ &\leq \int_{-\infty}^{\infty} \left|e^{itx}\right| dF(\boldsymbol{X}) \\ &= \int_{-\infty}^{\infty} dF(\boldsymbol{X}) = 1 \end{aligned}$$

Hence, characteristic function exists for arbitrary distribution.

Properties of characteristic function

(CF1) Moments can be obtained from derivatives of the characteristic function.

$$\varphi^{(m)}(t) = E\left\{(i\boldsymbol{X})^m e^{it\boldsymbol{X}}\right\}$$

$$\varphi^{(m)}(0) = E\{(i\boldsymbol{X})^m\}$$

Hence $$\mu'_m = E(\boldsymbol{X}^m) = \frac{\varphi^{(m)}(0)}{i^m}$$

$$E(\boldsymbol{X}) = \frac{\varphi'(0)}{i}$$

$$Var(\boldsymbol{X}) = \frac{\varphi''(0)}{i^2} - \left(\frac{\varphi'(0)}{i}\right)^2$$

(CF2) If there exist central moment lower than *m*-th order μ'_k $(1 \leq k \leq m)$, characteristic function $\varphi(t)$ can be expanded around $t = 0$ into Taylor series:

$$\varphi(t) = \sum_{k=0}^{\infty} \frac{1}{k!} \varphi^{(k)}(0) t^k = 1 + \sum_{k=1}^{m} \frac{(it)^k}{k!} \mu'_k + O(t^{m+1})$$

(CF3) Suppose random variables $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ are mutually independent and characteristic function of $\boldsymbol{X}_i$ is given by $\varphi_i(t)$. Then, characteristic function of the following variable

$$S = \boldsymbol{X}_1 + \boldsymbol{X}_2 + \cdots + \boldsymbol{X}_n$$

is given by

$$\varphi_S(t) = \varphi_1(t)\, \varphi_2(t) \cdots \varphi_n\,(t)$$

(CF4) Consider two random variables $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$. Suppose their distribution functions are given respectively by $F_1(x)$ and $F_2(x)$ and their characteristic functions are given by $\varphi_1(t)$ and $\varphi_2(t)$.

The necessary and sufficient condition of $F_1(x) = F_2(x)$ is $\varphi_1(t) = \varphi_2(t)$.

Probability density function is obtained by inversion formula.

- Discrete-type distribution function:

$$f(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikx}\varphi(t)dt$$

- Continuous-type distribution function:

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\varphi(t)dt$$

(CF5) (Continuity theorem)

Suppose $F_1(x), F_2(x), \cdots$ represent a sequence of distributions and $\varphi_1(t), \varphi_2(t), \cdots$ represent a sequence of corresponding characteristic functions. If the series $\varphi_1(t), \varphi_2(t), \cdots$ converges for any constant t

$$\lim_{n\to\infty} \varphi_n(t) = \varphi(t),$$

and the convergence value $\varphi(t)$ is continuous at $t = 0$, $\varphi(t)$ gives the characteristic function.

For a distribution function $F(x)$ that corresponds to $\varphi(t)$.

$$\lim_{n\to\infty} F_n(x) = F(x).$$

(CF6) Characteristic function for joint distribution of random variables $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ is defined by

$$\varphi(t_1, t_2, \cdots, t_n) = E\{\exp[i(t_1\boldsymbol{X}_1 + t_2\boldsymbol{X}_2 + \cdots t_n\boldsymbol{X}_n)]\}$$

Necessary and sufficient condition for mutual independency of $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ is given by

$$\varphi(t_1, t_2, \cdots, t_n) = \varphi_1(t_1)\varphi_2(t_2)\cdots\varphi_n(t_n).$$

Where:

$$\varphi_i(t_i) = \varphi_1(0, \cdots, 0, t_i, 0, \cdots, 0)\ .$$

12. Application of characteristic function

Law of Large Numbers

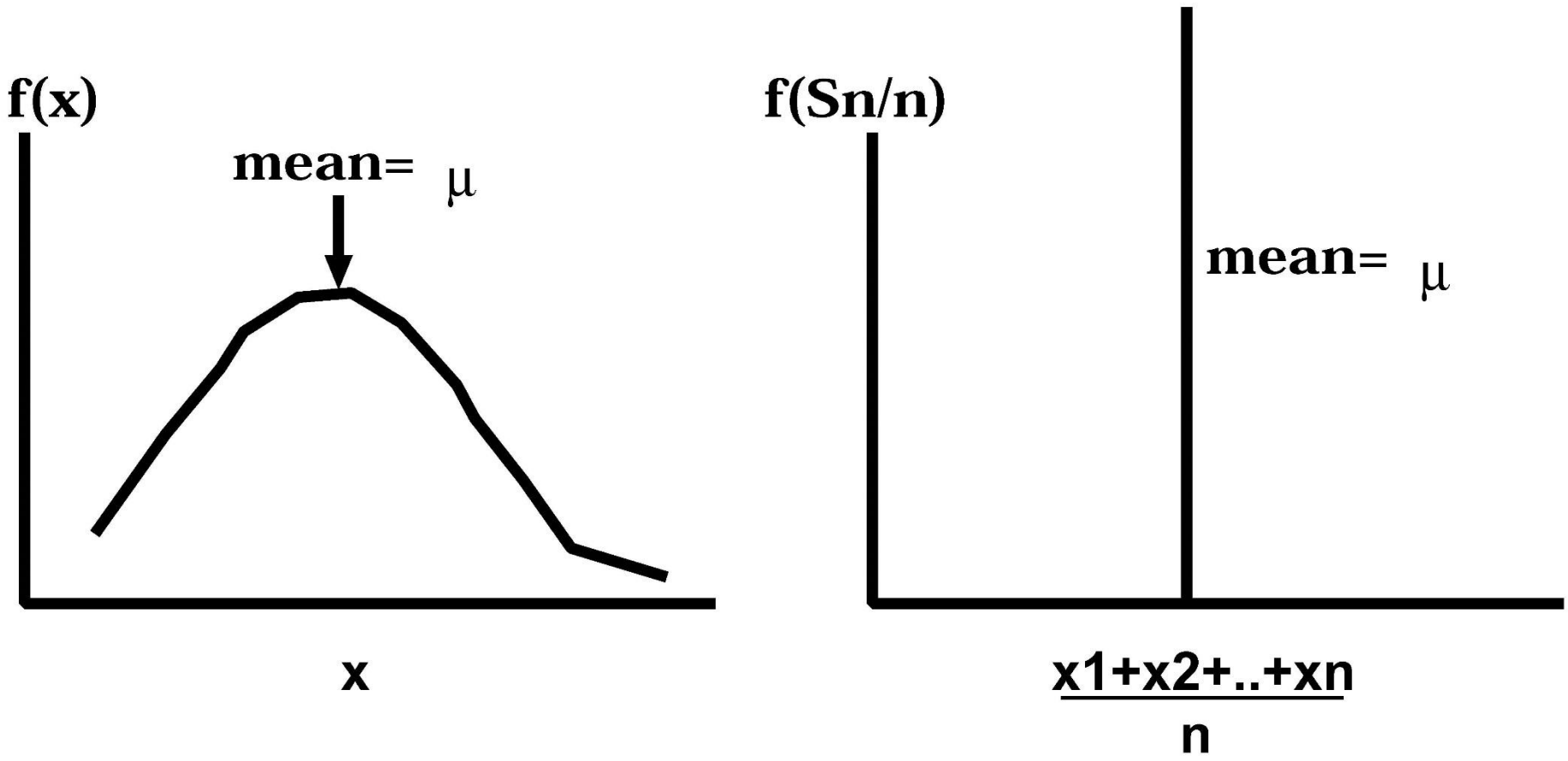
Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ be mutually independent random variables which obey the same distribution. Suppose that the distribution has a finite expectation value $\mu = E(\boldsymbol{X}_j)$, where its characteristic function is given by $\varphi(t)$.

Let

$$S_n = \boldsymbol{X}_1 + \boldsymbol{X}_2 + \cdots + \boldsymbol{X}_n$$

Then for any $\epsilon > 0$,

$$\lim_{n\to\infty} P\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) = 0$$


f(x)
mean= μ
x
f(Sn/n)
mean= μ
x1+x2+..+xn
n

- [Proof] Let characteristic functions of $\boldsymbol{X}_i$ and $\frac{S_n}{n}$ be $\varphi(t)$ and $\varphi_n(t)$, respectively. $\varphi_n(t) = \left[\varphi\left(\frac{t}{n}\right)\right]^n$

From (CF2), Taylor's expansion of $\varphi(t)$ around $t = 0$ yields $\varphi(t) = 1 + i\mu t + O(t^2)$. On the other hand, Taylor's expansion of $e^{i\mu t}$ around $t = 0$ gives $e^{i\mu t} = 1 + i\mu t + O(t^2)$.

Hence, $\varphi(t) = e^{i\mu t} + O(t^2)$. This leads to

$$\varphi_n(t) = \left[\varphi\left(\frac{t}{n}\right)\right]^n = \left[e^{\frac{i\mu t}{n}} + O(t^2)\right]^n \to e^{i\mu t} \ (n \to \infty)$$

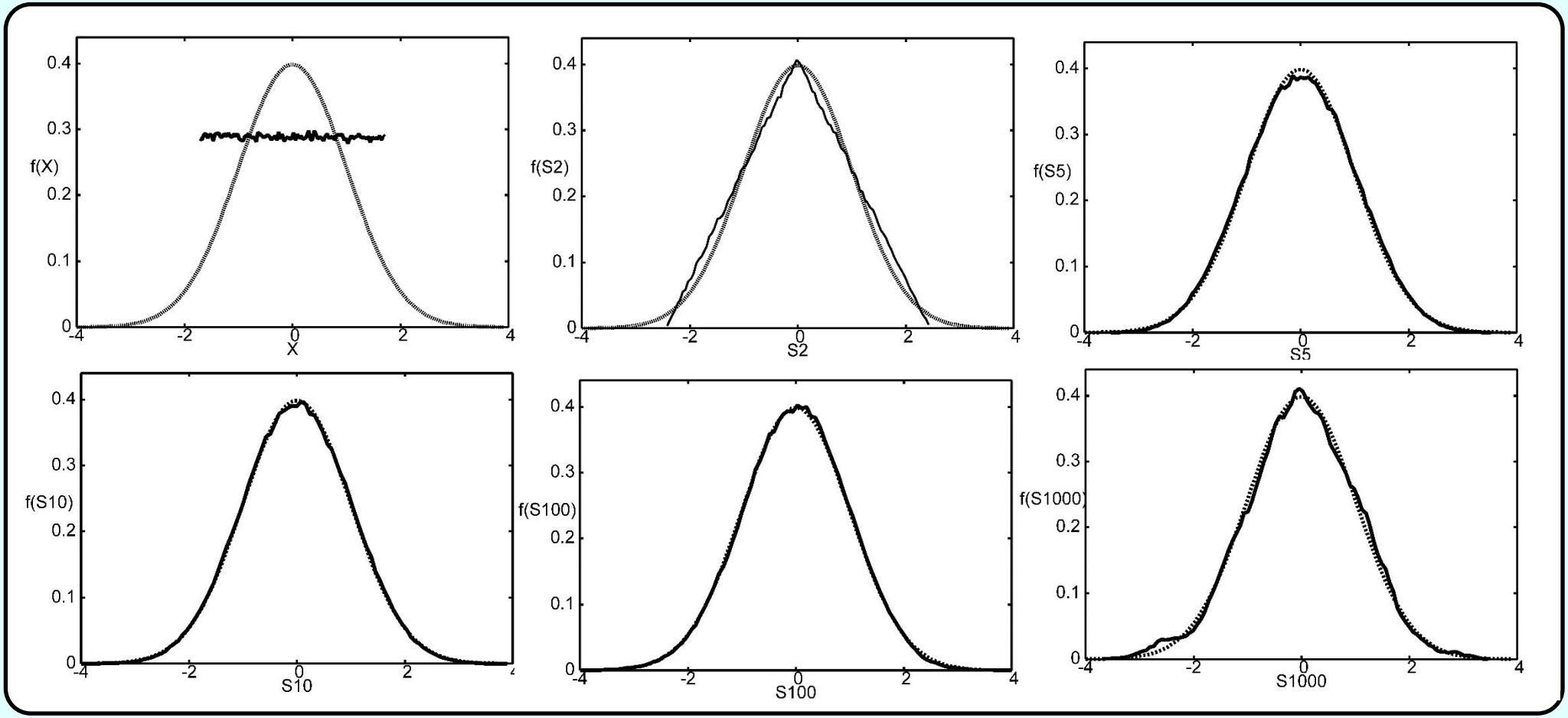
$e^{j\mu t}$ represents characteristic function of a random variable that takes a value of μ with a probability of 1. From the continuity theorem, $\varphi_n(t) \to e^{i\mu t}, \quad \frac{S_n}{n} \to \mu \quad (n \to \infty)$.

Central Limit Theorem

$\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_n$ be a sequence of *n* independent and identically distributed random variables, each having finite values of expectation μ and variance σ^2 (> 0).

Let S_n^* be

$$S_n^* = \sum_{j=1}^{n} \frac{X_j - \mu}{\sqrt{n}\sigma}.$$

Then, as *n* approaches infinity $(n \to \infty)$, S_n^* will converge in distribution to the standard normal distribution $N(0,1)$.

Example of central limit theorem: Summation of uniform random variables